An Introduction to Computational Stochastic PDEs, CUP, 2014

Here are the typos/errors that we know about in the first edition. Detailed corrections to MATLAB codes are given online[1]. Let us know if you find any more.
16th October 2026

Chapter 1. **p2** The statement that functions in $\mathrm{C}(\overline{D})$ are bounded, so that $(\mathrm{C}(\overline{D}), \|\cdot\|_\infty)$ is a normed vector space, requires D to be bounded. For $D = (0,\infty)$, the function $u(x) = x$ belongs to $\mathrm{C}([0,\infty))$ and $\|u\|_\infty = \infty$. The later definitions, which assume D bounded, are correct.

p7 In property (ii) of the integral, "For integrable $u\colon X \to Y$" (add *integrable*).

p8 In Figure 1.2(b), the assertion that $u(x) = x^{-2}$ belongs to $L^p(1,\infty)$ for $p > 2$ is not optimal. Since $\int_1^\infty x^{-2p}\,dx$ is finite whenever $p > 1/2$, we have $u \in L^p(1,\infty)$ for every $p \geq 1$.

p10 In the definition of $L^2(\Omega, L^2(D))$, the condition should read $u(\cdot,\omega) \in L^2(D)$ for almost every $\omega \in \Omega$ (not $\omega \in D$).

p13 In Definition 1.42, the weak derivative of a Y-valued function is also Y-valued, $\mathcal{D}^\alpha u\colon D \to Y$ (not $\mathbb{R}$-valued as printed). Further, for the integrals to exist, u and $\mathcal{D}^\alpha u$ should be integrable on every compact subset of D. Definition 1.44 should say "We say $\mathcal{D}^\alpha u \in L^p(D,Y)$ is…".

p14 In Definition 1.47(ii), $H^2_{\mathrm{per}}(a,b)$ is a Hilbert space with the $H^2(a,b)$ inner product (rather than the $H^2(D)$ inner product).

p17 In Definition 1.60, the Hilbert–Schmidt norm should be defined as

$$\|L\|_{\mathrm{HS}(U,H)} \coloneqq \left(\sum_{j=1}^{\infty} \|L\phi_j\|^2\right)^{1/2},$$

(the norm on the right is the one in H not in U).

p19 "Closed and bounded sets are not compact in infinite dimensions" should read "need not be compact" (finite sets are compact), and "integral operators map bounded sets to compact ones" should read "to sets with compact closure", as in Definition 1.67.

p20 In Theorem 1.73(ii), and again in the proof of Theorem 1.80 (p22), the eigenfunctions form an orthonormal basis for the *closure* of the range of L. The range of a compact operator of infinite rank is not closed; for the Dirichlet Laplacian of Example 1.83, the range of $L = A^{-1}$ is $\mathcal{D}(A)$, whereas the eigenfunctions are a basis of $L^2(0,1)$.

p21 Lemma 1.78 (Dini's lemma) requires additionally the assumption that f is continuous (to guarantee that $f(x) - f_n(x) \geq \epsilon$ for the limit point x). In the application of Dini's lemma for the proof of Theorem 1.80 (Mercer's theorem), this holds true for $g(x) = G(x,x)$.

p21 In the proof of Lemma 1.77, non-negative definiteness of L means $\langle u, Lu\rangle \geq 0$. Its approximation of G should use sets $F_j \in \mathcal{B}(D)$ and the same points $\boldsymbol{x}_j$ in both arguments. For u vanishing outside a compact $K \subset D$, partition K into sets $F_j \ni \boldsymbol{x}_j$ of small diameter, so that $G_N \coloneqq \sum_{j,k} G(\boldsymbol{x}_j,\boldsymbol{x}_k)1_{F_j}(\boldsymbol{x})1_{F_k}(\boldsymbol{y})$ converges to G uniformly on $K \times K$. Then $\int_K \int_K u(\boldsymbol{x})G_N(\boldsymbol{x},\boldsymbol{y})u(\boldsymbol{y})\,d\boldsymbol{x}\,d\boldsymbol{y} = \sum_{j,k} a_j a_k G(\boldsymbol{x}_j,\boldsymbol{x}_k) \geq 0$ with $a_j = \int_{F_j} u$, and the error is at most $|K| \sup_{K\times K} |G - G_N|\,\|u\|^2$. The general case $u \in L^2(D)$ follows by approximating u by compactly supported functions, since $u \mapsto \langle u, Lu\rangle$ is continuous.

p22 In the proof of Theorem 1.80 (Mercer's theorem), two steps need justification.

- "$G - G_N$ is non-negative definite, as the corresponding integral operator has non-negative eigenvalues" uses the converse of Lemma 1.77: a continuous kernel whose integral operator is non-negative definite is a non-negative definite function. This follows by taking $u = \sum_j a_j 1_{B_\epsilon(\boldsymbol{x}_j)}/|B_\epsilon|$ and letting $\epsilon \to 0$.
- Dini's lemma needs the limit g to be continuous (see the entry for p21). For fixed $\boldsymbol{x}$, the Cauchy–Schwarz bound in the proof shows that $\sum_j \lambda_j \phi_j(\boldsymbol{x})\phi_j(\boldsymbol{y})$ converges uniformly in $\boldsymbol{y}$, so the limit is continuous in $\boldsymbol{y}$. It equals $G(\boldsymbol{x},\cdot)$ in $L^2(D)$, and so everywhere, and hence $g(\boldsymbol{x}) = \sum_j \lambda_j \phi_j(\boldsymbol{x})^2 = G(\boldsymbol{x},\boldsymbol{x})$ is continuous.

[1] `https://github.com/tonyshardlow/picspde`

p23, p24 In Examples 1.83 and 1.84, the eigenvalue problems are printed with $\frac{d\phi_j^2}{dx^2}$ and $\frac{d\phi^2}{dx^2}$, which should read $\frac{d^2\phi_j}{dx^2}$ and $\frac{d^2\phi}{dx^2}$.

p27 Proposition 1.93 holds only for periodic functions: the norm of $H^r(-\pi,\pi)$ is equivalent to $\|u\|_{r/2}$ for $u \in \mathcal{D}(A^{r/2})$, and the proof should start "For $u \in \mathcal{D}(A^{r/2})$".

p29 In (1.33), the first identity requires the conjugate transpose (missing conjugation) and should read

$$\widehat{\boldsymbol{u}}^{\,*}\widehat{\boldsymbol{v}} = d\,\boldsymbol{u}^{\,*}\boldsymbol{v},$$

where * denotes the conjugate transpose.

p29 For real data $\boldsymbol{u} \in \mathbb{R}^d$, the symmetry of the DFT should read $\hat{u}_\ell = \overline{\hat{u}_{d+2-\ell}}$ (indices modulo d), not $\hat{u}_\ell = \overline{\hat{u}_{d-\ell}}$. The latter is correct only under the convention $\ell = 0, \dots, d-1$, whereas the book indexes $\ell = 1, \dots, d$.

p32 In the example following Lemma 1.100, the sentence should read that $\|I_J u\|_{L^2}$ and $\|I_J u\|_{H^1}$ provide estimates of $\|u\|_{L^2}$ and $\|u\|_{H^1}$ (not of $\|u\|_{H^1}$ and $\|u\|_{H^2}$).

p32 In Lemma 1.100, the bound holds with K/J in place of $K/J^{1/2}$. The stated bound is true but not sharp. See Exercise 1.23(b).

p32 In the example following Lemma 1.100, the second display should read

$$\|I_J u\|_{H^1(a,b)} = \Big(\|I_J u\|^2_{L^2(a,b)} + (b-a) \sum_{k=-J/2+1}^{J/2} \frac{(2\pi k)^2}{(b-a)^2} |U_k|^2 \Big)^{1/2},$$

with $\|I_J u\|_{L^2}$, not the unknown $\|u\|_{L^2}$, inside the bracket.

p34 In Theorem 1.107 (Hankel transform), the constant in the case $d = 3$ should be $\sqrt{2/\pi}$ and not $\sqrt{2\pi}$:

$$\hat{u}(s) = \sqrt{\frac{2}{\pi}} \int_0^\infty \frac{\sin(rs)}{rs}\, u(r)\, r^2\, dr.$$

This follows from the proof, which gives $\int_{\mathbb{R}^3} \mathrm{e}^{-\mathrm{i}\lambda^{\mathsf{T}}\boldsymbol{x}} u(\boldsymbol{x})\, d\boldsymbol{x} = 4\pi \int_0^\infty \frac{\sin(rs)}{rs} u(r) r^2\, dr$, together with $4\pi/(2\pi)^{3/2} = \sqrt{2/\pi}$. It is also consistent with the case $d = 1$.

p34 In Example 1.105, the sentence should read that the Fourier transform of the derivative equals $\mathrm{i}\lambda$ times the Fourier transform (not "the derivative of the Fourier transform"), in agreement with the preceding display $\widehat{u'}(\lambda) = \mathrm{i}\lambda\hat{u}(\lambda)$.

p38 In Exercise 1.23, the interval should be $(0,1)$ throughout, not $(0, 2\pi)$. Then P_J projects onto $\operatorname{span}\{\mathrm{e}^{2\pi \mathrm{i} j x} : j = -J/2+1, \dots, J/2\}$ as printed, and u_1 of Figure 1.6 is defined on this interval.

In Part (c), the final rate should read

$$\|u_1\|_{H^1(0,1)} - \|I_J u_1\|_{H^1(0,1)} = \mathcal{O}(J^{-4}),$$

(the printed bound is correct but sub-optimal).

The first bound in Part (c) should require $0 < 2r < p$, not $0 < 2r \le p$. At $2r = p$, the aliasing sum is of order $\log J$ and the rate becomes $\mathcal{O}(J^{-(1+p)} \log J)$.

p39 In Exercise 1.28, "Making use of DCT-1, evaluate $u^N(x)$ at $x = (k-1)\pi/(N-1)$" should read $v^N(x)$, the function defined immediately above and plotted in the same sentence.

Chapter 2. **p41** In Example 2.3, the formula for the solution requires $p, q > 0$: "If p, q and f are constants with $p, q > 0$". It divides by q and involves $\sqrt{q/p}$, whereas Assumption 2.1 allows $q = 0$.

p42 Figure 2.1 is misprinted.

p42 In the derivation of the weak form, the test function should be $\phi \in \mathrm{C}_c^\infty(a,b)$, not $\mathrm{C}_c^\infty([a,b])$. Functions in $\mathrm{C}_c^\infty([a,b])$ need not vanish at a and b, and $\phi(a) = \phi(b) = 0$ is used immediately afterwards.

p44 ℓ is a bounded linear *functional* on $H_0^1(a,b)$, not a bounded linear operator, as in the Lax–Milgram lemma.

p47 The Galerkin projection P_G is the orthogonal projection onto $\tilde{V}$ with respect to the energy inner product $a(\cdot,\cdot)$, not the $H_0^1(a,b)$ inner product; this is what the best-approximation property in the energy norm expresses.

p48 In the proof of Lemma 2.20, the eigenfunctions form an orthonormal basis for G, the *closure* of the range of L (see the entry for p20), which need not be all of $L^2(a,b)$. By Theorem 1.37, every $u \in L^2(a,b)$ is $u = p + q$ with $p \in G$ and $q \in G^\perp$, so it suffices to show $G^\perp = \{0\}$. If $f \in G^\perp$, then $\langle f, \phi_j \rangle = 0$ for all j, so $Lf = 0$ by (1.21) and hence $f = ALf = 0$.

p49 In Example 2.22, the odd-periodic extension of f must be analytic (rather than f itself).

p56 In the first displayed equation of the proof, the first term in the integrand should be $p(x)(e(x)')^2$ (the $e(x)$ has one too many dashes).

p57 In the proof of Theorem 2.32, the step to (2.44) uses $h^4 \le h^2$, that is $h \le 1$. On a general interval (a,b), either assume $h \le 1$ or replace $\|q\|_\infty/\pi^4$ by $(b-a)^2\|q\|_\infty/\pi^4$.

p62 In the proof of Theorem 2.43, the displayed bound following (2.60) should read

$$|u_0|_{H^1(D)} \le a_{\min}^{-1}\Big(K_{\mathrm{p}}\|f\|_{L^2(D)} + a_{\max}K_\gamma\|g\|_{H^{1/2}(\partial D)}\Big),$$

with the constant K_γ present in the second term. This is what (2.60) gives on dividing by $|u_0|_{H^1(D)}$, and it is needed for the constant $K \coloneqq \max\{K_{\mathrm{p}}a_{\min}^{-1},\, K_\gamma(1 + a_{\max}a_{\min}^{-1})\}$ stated in the theorem: combining it with $|u_g|_{H^1(D)} \le K_\gamma\|g\|_{H^{1/2}(\partial D)}$ from (2.54) reproduces exactly that K.

p79 The right-hand side of (2.109) should be $c_1|\hat{u} - I_h\hat{u}|^2_{H^2(\Delta^*)}$.

Assumption 2.64 in general can only be verified for constant boundary data g; H^2-regularity results usually include an extra term on the right-hand side to account for $g \neq 0$ – see (Renardy and Rogers, 2004, Theorem 8.53) or (McLean, 2000, Theorem 4.10).

p79 "u is not guaranteed to be continuous and so neither is $I_h u$": the interpolant $I_h u$ is piecewise linear and continuous by construction. The difficulty is that the point values of u used in (2.105) need not be defined.

p83 In Exercise 2.4(a), the jump (2.114) is taken across $x = y$ and is better written

$$\big[G'(x,y)\big]_{x=y_-}^{x=y_+} = \frac{-1}{p(y)},$$

that is, the limit of $G'(y+\epsilon, y) - G'(y-\epsilon, y)$ as ϵ approaches 0 from above. As printed, G' denotes $\partial G/\partial x$ but the two limits are taken in the second argument, so the sentence does not describe the quantity being computed: the derivation integrates the equation over $x \in [y-\epsilon, y+\epsilon]$ at fixed y. Symmetry of G makes the two expressions agree numerically, so nothing that follows is affected.

p83 In Exercise 2.6, the element matrix should be $\frac{p}{3h}\begin{pmatrix} 7 & -8 & 1 \\ -8 & 16 & -8 \\ 1 & -8 & 7 \end{pmatrix}$, with the factor p; equivalently, take $p = 1$.

Chapter 3. **p98** In Example 3.13, the Fourier transform of u_x should be $\widehat{u_x}(\lambda) = \mathrm{i}\lambda\widehat{u}$ and not $-\mathrm{i}\lambda\widehat{u}$. This is the identity derived in Example 1.105, from which it is quoted. Equation (3.25) is unaffected, since the symbol is squared.

p98 In Example 3.14, $\mathrm{C}(\mathbb{R})$ with the supremum norm is not a Banach space, since it contains unbounded functions. The semigroup should be defined on the bounded, uniformly continuous functions, as in the Notes (p132).

p99 In Example 3.16, $\hat{u}_0(\lambda)$ is complex-valued, and the display for $\|u(t) - u_0\|$ should contain $|\hat{u}_0(\lambda)|^2$ in place of $\hat{u}_0(\lambda)^2$.

p101 The proof of Lemma 3.21 shows only that the generator G of e^{-tA} extends $-A$, that is $\mathcal{D}(A) \subset \mathcal{D}(G)$ and $Gu = -Au$ there. For the reverse inclusion, let $u \in \mathcal{D}(G)$ and $w = Gu$. Taking the inner product of $(\mathrm{e}^{-hA}u - u)/h$ with ϕ_j and letting $h \downarrow 0$ gives $\langle w, \phi_j\rangle = -\lambda_j\langle u, \phi_j\rangle$. So $\sum_j \lambda_j^2\langle u, \phi_j\rangle^2 = \|w\|^2 < \infty$ and $u \in \mathcal{D}(A)$.

p105 In the proof of Lemma 3.30, the displayed bound should read

$$\int_D \tilde{f}(u(\boldsymbol{x}))^2\,d\boldsymbol{x} \le (L')^2\int_D \big(1 + |u(\boldsymbol{x})|\big)^2\,d\boldsymbol{x},$$

with the modulus of u and with $L' = \max\{L, |\tilde{f}(0)|\}$ in place of the Lipschitz constant L. As printed, with $(1 + u(\boldsymbol{x}))^2$, the inequality fails for negative u: for $\tilde{f}(r) = r$ (Lipschitz with $L = 1$) and $u \equiv -1$ the left-hand side is $|D|$ and the right-hand side is 0.

p105 Proposition 3.31 requires $\epsilon \in (0, 1]$ (the proof applies Lemma 3.22(c) with $\alpha = 1 - \epsilon$), and its proof uses $t_2 - t_1 \le 1$ when it combines the terms. For $t_2 - t_1$ up to T, the constant should carry a factor such as $(1 + T^{\epsilon})$.

p107 In (3.46) the remainder term should be $\varepsilon\, r_j(t)$, and correspondingly the last term of (3.47) should be $\varepsilon\, \boldsymbol{r}(t)$. The remainder r_j is defined in (3.45) as that of the difference approximation of u_{xx}, which is multiplied by ε on substitution into (3.44). The $O(h^2)$ conclusion is unchanged.

p107 Taylor's theorem is applied to $u(t, \cdot) \in \mathrm{C}^4([0, a])$; the problem is posed on $(0, a)$, not $\mathbb{R}$.

p115 In (3.71) and (3.72), the factor should be $(1 + \Delta t\, \lambda_j)^{-1}$ and not $(1 + \Delta t\, \varepsilon\lambda_j)^{-1}$. The eigenvalues are defined in Example 3.39 as $\lambda_j = \varepsilon j^2$, so the ε is already included; as printed the factor is $(1 + \Delta t\, \varepsilon^2 j^2)^{-1}$. Compare the general scheme (3.68), the system (3.70), and Algorithm 3.5, all of which carry a single ε.

p116 In Example 3.40, Figure 3.5(b) was computed not from the stated initial data $u_0 = \sin(x_1)\cos(\pi x_2/8)$ but from $\sin(x_2)\cos(\pi x_1/8)$, which jumps by 1.78 and 0.29 across the two periodic boundaries and so cannot have been intended.

p116 In Example 3.40, the eigenvalues are likewise defined to include the diffusion coefficient, $\lambda_{j_1,j_2} = \varepsilon((2\pi j_1/a_1)^2 + (2\pi j_2/a_2)^2)$. The right-hand side of (3.73) should therefore be $-\lambda_{j_1,j_2}\widehat{u}_{j_1,j_2} + \widehat{f}_{j_1,j_2}$, and the entries of M should be $m_{i,j} = \lambda_{j_1,j_2}$, in both cases without the additional ε. Algorithm 3.6 uses a single ε.

p118 In the proof of Lemma 3.42, the factor inside the modulus should read $(1 + x)^{-1} - \mathrm{e}^{-x}$, which is non-negative, rather than $\mathrm{e}^{-x} - (1 + x)^{-1}$.

p119 In the caption of Algorithm 3.7, the spacing of the returned grid should be $h_{\mathrm{ref}} = a/J_{\mathrm{ref}}$ and not $1/J_{\mathrm{ref}}$, the domain being $D = (0, a)$.

p123 In Example 3.45, $T = 1$ and $N = 10^4$ give $\Delta t = 10^{-4}$, not 0.001. The computations use $\Delta t = 10^{-4}$.

p126 In Example 3.49, the error is defined as $e \coloneqq u_h - u$ but used later as $e = u - u_h$; define $e \coloneqq u - u_h$ throughout.

p127 In the proof of Lemma 3.51, the second application of Young's inequality should read

$$2s\left\|\frac{d\rho}{ds}\right\| \|e(s)\| \le s^2\left\|\frac{d\rho}{ds}\right\|^2 + \|e(s)\|^2,$$

the right-hand side carrying the same arguments as the left; as printed it reads $s^2\|\rho(t)\|^2 + \|e(t)\|^2$. The display that follows carries the correct terms and the conclusion is unaffected.

p127 In the proof of Lemma 3.51, multiplying by t gives $\frac{d}{dt}(t\|e\|^2) = t\frac{d}{dt}\|e\|^2 + \|e\|^2$, so a term $+\|e(t)\|^2$ is missing from the differential inequality and from (3.93). Carrying it through gives $3\|e(s)\|^2$ and then $4\|\rho(s)\|^2$ in the next two displays, and the constant in the lemma becomes $\sqrt{12}$ in place of $\sqrt{10}$.

p128 The proof of Proposition 3.52 is valid only for $\gamma = 1$. For $\gamma < 1$, the bounds on $\|\rho(s)\|$ and $s\|d\rho/ds\|$ blow up as $s \to 0$, so the supremum in Lemma 3.51 is infinite.

- For $\gamma \in [0, 1]$, (3.94) is true and follows from the t-weighted energy estimate

$$\|e(t)\| \le \frac{15}{t}\sup_{0<s\le t}\left(\|\hat{\rho}(s)\| + s\|\rho(s)\| + s^2\left\|\frac{d\rho}{ds}(s)\right\|\right), \qquad \hat{\rho}(t) \coloneqq \int_0^t \rho(s)\, ds,$$

  (Thomée, *Galerkin Finite Element Methods for Parabolic Problems*, 2006, Lemma 3.4) together with $\hat{\rho}(t) = (A^{-1} - \tilde{A}^{-1})(I - \mathrm{e}^{-tA})u_0$ and Lemma 3.22(c).

- For $\gamma = -\alpha < 0$ and finite elements, (3.94) is false. Write $e(t) = E(t)u_0$. The operator $E(t)$ is self-adjoint on H, so $\sup_{\|u_0\|_{-1/2}\le 1}\|e(t)\| = \|A^{1/2}E(t)\|_{\mathcal{L}(H)}$, which is the $H_0^1(D)$ error for L^2 data. For piecewise linear elements this is of order h, not h^2. The correct bound for $\alpha \in [0, 1/2]$ is

$$\|e(t)\| \le K\frac{\delta^{2-2\alpha}}{t}\|u_0\|_{-\alpha}.$$

  The original bound $\delta^2 t^{-(1+\alpha)}\|u_0\|_{-\alpha}$ does hold for the spectral Galerkin method.

p128 In the proof of Proposition 3.53, the bound $\|\tilde{P}u\| \le C^{\alpha}\delta^{-2\alpha}\|u\|_{-\alpha}$ does not follow from (3.86) and Exercise 3.20. These give $\|v\| \le C^{\alpha}\delta^{-2\alpha}\|v\|_{-\alpha}$ only for $v \in \tilde{V}$, and nothing bounds $\|\tilde{P}u\|_{-\alpha}$ by $\|u\|_{-\alpha}$. Assume instead the inverse estimate $\|A^{1/2}v\| \le C\delta^{-1}\|v\|$ for $v \in \tilde{V}$, which holds for the spectral Galerkin method and for finite elements on quasi-uniform meshes (see, for example, Brenner and Scott,

The Mathematical Theory of Finite Element Methods, §4.5). For $\alpha \in [0, 1/2]$, Exercise 3.20 gives $\|A^\alpha v\| \le C^{2\alpha}\delta^{-2\alpha}\|v\|$ on $\tilde V$ and, as $\tilde P$ is the orthogonal projection,

$$\|\tilde P u\| = \sup_{v\in\tilde V} \frac{\langle A^{-\alpha}u, A^\alpha v\rangle}{\|v\|} \le C^{2\alpha}\delta^{-2\alpha}\|u\|_{-\alpha}.$$

So (3.96), and (3.98) in Theorem 3.54, hold for $\alpha \in [0, 1/2]$.

Separately, the extension of (3.95) to $\gamma \in (0, 1)$ cannot apply Exercise 3.20 to $T_n\tilde P$ directly, as $T_n\tilde P$ does not commute with A (see the entry for p136). Instead, by duality, $\|T_n\tilde P A^{-\gamma}\| = \|A^{-\gamma}(T_n\tilde P)^*\|$, and the first inequality of Exercise 3.20, with A^{-1} in place of A and applied to $(T_n\tilde P)^* w$, gives $\|T_n\tilde P A^{-\gamma}\| \le \|T_n\tilde P A^{-1}\|^\gamma \|T_n\tilde P\|^{1-\gamma}$. This inequality is also Theorem 6.10 in §2.6 of Pazy, *Semigroups of Linear Operators and Applications to Partial Differential Equations*, 1983.

p129 In Theorem 3.54, the quantifier in (3.98) should read $\forall u_0 \in \mathcal{D}(A^{-\alpha})$ and not $\forall u_0 \in \mathcal{D}(A^\alpha)$. The bound is stated in terms of $\|u_0\|_{-\alpha}$ and follows by combining (3.94) and (3.96), each stated for $u_0 \in \mathcal{D}(A^{-\alpha})$. The distinction matters: as noted on p128, $\mathcal{D}(A^{-\alpha})$ admits data that need not belong to H, and it is that case which is required for space–time white noise forcing later in the book.

(3.98) should be stated for $\alpha \in [0, 1/2]$, and for finite elements its second term $\delta^2/t_n^{1+\alpha}$ should read $\delta^{2-2\alpha}/t_n$ (see the entry for p128). The application in Chapter 10, with $\Delta t = \delta^2$ and $\alpha = (1-\zeta)/2$, gives the same rate Δt^ζ with the corrected bound.

p132 Following the comment on p487 below, the last line of the proof should be

$$\|u(t_n) - \tilde u_n\| \le E\ \exp(Ln\Delta t).$$

p132 In the Notes, the mapping $t \mapsto S(t)u$ of an analytic semigroup is analytic on $(0, \infty)$, and not on $[0, \infty)$, for $u \in X$. Analyticity at $t = 0$ would require $u \in \mathcal{D}(A^n)$ for every n; for general $u \in X$ the mapping is only strongly continuous there.

p132 In the Notes, a sectorial operator is one whose *spectrum* lies in a sector of the complex plane, with a resolvent bound outside it, not merely one whose eigenvalues do.

p134 Exercise 3.4 requires $b > 0$. For $b < 0$ the stated conclusion is false: take $a = 0$, $b = -1$, $z_0 = 1$, and $z(t) = 1/10$ for $t \in [0, 9]$, $z(t) = \mathrm{e}^{-(t-9)}/10$ for $t \ge 9$. Then z satisfies (3.102) for all $t \ge 0$, but $z(5) = 1/10 > \mathrm{e}^{-5}$, whereas the claimed bound is $\mathrm{e}^{bt}z_0 + (a/b)(\mathrm{e}^{bt} - 1) = \mathrm{e}^{-5}$.

p135 In Exercise 3.9, the step sizes should be related by $\Delta t_0 = \kappa\,\Delta t_1$, so that $\boldsymbol{u}_{N_1}$ is the finer approximation. With $\Delta t_1 = \kappa\,\Delta t_0$ as printed, eliminating C_1 gives

$$\frac{\kappa\boldsymbol{u}_{N_1} - \boldsymbol{u}_{N_0}}{\kappa - 1} = \boldsymbol{u}(T) - C_1(\kappa + 1)\Delta t_0 + O(\Delta t_0^2),$$

so the $O(\Delta t_0)$ term does not cancel. The corrected relation also matches Exercise 3.10, where $\Delta t_1 = 0.01$ and $\Delta t_0 = 0.1$.

p135 In Exercise 3.11, $H_0^1(D) \times L^2(D)$ is the space in which the first-order system evolves, not the domain of the generator; the phrase should read "in the space $H = H_0^1(D) \times L^2(D)$". The domain is the smaller set $\mathcal{D}(A) = (H^2(D) \cap H_0^1(D)) \times H_0^1(D)$.

p135 In Exercise 3.11, the first-order system is a *linear* equation $d\boldsymbol{u}/dt = -A\boldsymbol{u}$, not a semilinear one.

p136 In Exercise 3.20, the second inequality requires that B commute with A (for instance, that B be a function of A); it is false for general $B \in \mathcal{L}(H)$.

Chapter 4. **p145** "Uncorrelated random variables are not independent in general, except in the special case of Gaussian random variables" is false as stated: two uncorrelated Gaussian random variables need not be independent. Take $X \sim \mathrm{N}(0, 1)$ and $Y = SX$ with $S = \pm 1$ equally likely and independent of X; then $Y \sim \mathrm{N}(0, 1)$ and $\mathrm{Cov}(X, Y) = 0$, but $|Y| = |X|$. It should read "... except when they are the components of a multivariate Gaussian random variable (see Lemma 4.33)".

p156 In Example 4.51 (conditional Gaussian), the normalising constant of the pdf of $\boldsymbol{X}$ should read $1/\sqrt{(2\pi)^{n+m}\det Q}$, and each exponent should carry a factor $1/2$, in accordance with (4.13); the first display should read

$$p_{\boldsymbol{X}}(\boldsymbol{x}_1, \boldsymbol{x}_2) = \frac{1}{\sqrt{(2\pi)^{n+m}\det Q}}\,\mathrm{e}^{-(\boldsymbol{x}-\boldsymbol{\mu})^{\mathsf T}Q^{-1}(\boldsymbol{x}-\boldsymbol{\mu})/2},$$

and similarly for the two displays that follow. As printed, the final display makes the conditional density of $\boldsymbol{X}_2$ given $\boldsymbol{X}_1 = \boldsymbol{y}$ proportional to $\mathrm{e}^{-(\boldsymbol{x}_2-\widehat{\boldsymbol{\mu}}_2)^{\mathsf T}\overline{Q}_{22}^{-1}(\boldsymbol{x}_2-\widehat{\boldsymbol{\mu}}_2)}$, which is the density of $\mathrm{N}(\widehat{\boldsymbol{\mu}}_2, \overline{Q}_{22}/2)$ and not $\mathrm{N}(\widehat{\boldsymbol{\mu}}_2, \overline{Q}_{22})$ as concluded.

p156 In Example 4.51, the conditioned vector is $\boldsymbol{X}_2$, not $\boldsymbol{X}$: the display should read $\mathbb{P}(\boldsymbol{X}_2 \in G \mid \boldsymbol{X}_1 = \boldsymbol{y}) = \frac{1}{p_{\boldsymbol{X}_1}(\boldsymbol{y})} \int_G p_{\boldsymbol{X}}(\boldsymbol{y}, \boldsymbol{x}_2)\, d\boldsymbol{x}_2$, with $G \subset \mathbb{R}^m$, and the next sentence should refer to "the density of $\boldsymbol{X}_2$ given $\boldsymbol{X}_1 = \boldsymbol{y}$".

p159 Nensen's inequality → Jensen's inequality (in proof of Theorem 4.58(iii)).

p159 In the proof of Lemma 4.57 (Borel–Cantelli), the last sentence should read: for $\omega \notin E$, $|X_n(\omega)| > 1$ for only finitely many n, so $|X_n(\omega)| \le 1$ for all $n \ge n^*(\omega)$; since $\mathbb{P}(E) = 0$, (4.22) holds. The assertion $\mathbb{P}(n^* = 1) = 0$ does not follow in general.

p163 Theorem 4.63 holds only for $\epsilon < \lambda_{\min}$, the smallest eigenvalue of C_X, not for every $\epsilon > 0$. For larger ϵ, the ϵ-ball about C_X contains positive-definite C_Y arbitrarily close to singular, and no uniform L exists. With $\epsilon < \lambda_{\min}$, every admissible $C_Y \ge (\lambda_{\min} - \epsilon) I$ is positive definite, and the second-order remainder in the proof is bounded uniformly, which is what the choice of L needs.

p168 In the caption of Algorithm 4.7, `M` is the number of trials required to obtain `X`, and not the number of rejections: the counter is incremented on the accepted draw as well, so the number of rejections is `M-1`.

p174 In the first display of the Monte Carlo error analysis, $\overline{X}_N$ should read $\overline{X}_M$, the sample average being taken over the M Monte Carlo samples (N is the number of time steps).

p175 The quasi-Monte Carlo rate is misstated. "The error is reduced from $O(M^{-1/2})$ to $O(M^{-2})$ as shown in Exercise 4.24" should read "to $O(M^{-n})$ for every n when $M = 3^p$". Exercise 4.24 proves this stronger rate, but only for $M = 3^p$. For general M, the van der Corput sequence gives only $O(M^{-1} \log M)$, so the restriction to $M = 3^p$ is essential.

p178 In Exercise 4.5, the moments should read $\mathbb{E}\big[|X|^k\big] = \infty$ for $k \ge 1$. For $X \sim \text{Cauchy}(0, 1)$ and k odd, $\mathbb{E}[X^k]$ is undefined rather than infinite, since $\mathbb{E}\big[(X^k)^+\big] = \mathbb{E}\big[(X^k)^-\big] = \infty$.

p179 In Exercise 4.17, ϵ must satisfy $0 < \epsilon < \sigma_X^2$, for the reason given for Theorem 4.63 (p163): otherwise σ_Y^2 may approach 0 and no uniform L exists.

p179 Bayes' theorem is due to the Reverend Thomas Bayes and the apostrophe is written after and not, as in Exercise 4.11, before the s. In the same exercise, $p_{X,Y}$ is incorrectly defined and it should be

$$\mathbb{P}(X = x_k, Y = y_j) = P_{X,Y}(k, j)$$

(interchange j and k).

Chapter 5. **p184** In the proof of Lemma 5.8, the increment should read

$$X_{\lfloor t_{i+1} N \rfloor} - X_{\lfloor t_i N \rfloor} = \sum_{j=\lfloor t_i N \rfloor + 1}^{\lfloor t_{i+1} N \rfloor} \xi_{j-1},$$

to match the convention $X_n = \sum_{j=1}^{n} \xi_{j-1}$ used in the proof of Lemma 5.5.

p185 In the proof of Lemma 5.8, the first term of (5.6), $N^{-1/2}[X_{\lfloor t_1 N \rfloor}, \ldots, X_{\lfloor t_M N \rfloor}]^{\mathsf{T}}$, is not an affine transformation of the $M - 1$ increments (5.5) with $A \in \mathbb{R}^{M \times (M-1)}$: $X_{\lfloor t_1 N \rfloor}$ cannot be recovered from increments that start at t_1. Set $t_0 = 0$ and include the first increment $X_{\lfloor t_1 N \rfloor} - X_0$, so that (5.5) has M independent components; then $A \in \mathbb{R}^{M \times M}$ is the lower-triangular matrix of ones, and the rest of the proof is unchanged.

p188 In (5.12), the exponent should read $\mathrm{e}^{-(x - x_j)^2 / 2 t_n}$, with a minus sign, as in (5.13).

p200 In Example 5.26, the covariance function should read $C(s, t) = \big(|t|^{2H} + |s|^{2H} - |t - s|^{2H}\big)/2$, matching the definition of fBm in §5.3.2 and the factor `0.5` used in Algorithm 5.3 (`fbm.m`). The stated conclusion that C_N is singular is unaffected, since $C(0, t) = 0$ for either version.

p201 The display bounding $|\mathbb{E}[\phi(\boldsymbol{X})] - \mathbb{E}[\phi(\widehat{\boldsymbol{X}})]|$ by a multiple of $\|\phi\|_{L^2(\mathbb{R}^N)}$ is not justified, and no bound of that form holds: Theorem 4.63 needs both covariance matrices positive definite (with $\epsilon < \lambda_{\min}(C_N)$, see p163), whereas C_n has rank $n < N$. As $\widehat{\boldsymbol{X}}$ lives on an n-dimensional subspace, a narrow bump ϕ centred there has small L^2 norm but $\mathbb{E}[\phi(\widehat{\boldsymbol{X}})]$ of order one. For ϕ with bounded second derivatives, write $\boldsymbol{X} = \widehat{\boldsymbol{X}} + \boldsymbol{Z}$ with $\boldsymbol{Z} = \sum_{j=n+1}^{N} \sqrt{\nu_j}\, \boldsymbol{u}_j \xi_j$ independent of $\widehat{\boldsymbol{X}}$ and of mean zero; Taylor's theorem then gives

$$|\mathbb{E}[\phi(\boldsymbol{X})] - \mathbb{E}[\phi(\widehat{\boldsymbol{X}})]| \le \tfrac{1}{2} \sup_{\boldsymbol{x}} \|D^2 \phi(\boldsymbol{x})\|_2 \sum_{j=n+1}^{N} \nu_j,$$

the same order as the mean-square error.

p202 The statement "If the process is Gaussian" in Theorem 5.28 admits two readings. In the sense of Definition 4.38 (H-valued Gaussian, with $H = L^2(\mathcal{T})$), X Gaussian means $\langle X, \phi\rangle$ is a real-valued Gaussian random variable for every $\phi \in H$. Since $\xi_j = \langle X - \mu, \phi_j\rangle_{L^2(\mathcal{T})}/\sqrt{\nu_j}$ is of this form, and so is any linear combination $\sum_j \theta_j \xi_j = \langle X - \mu, \sum_j \theta_j \phi_j/\sqrt{\nu_j}\rangle$, the ξ_j are then immediately (jointly) Gaussian.

If instead "Gaussian" is read in the sense of Definition 5.10 (Gaussian finite-dimensional distributions), the claim needs justification, since $\gamma_j = \langle X - \mu, \phi_j\rangle_{L^2(\mathcal{T})}$ is an integral of X. It suffices that every linear combination $S = \sum_{j=1}^n \theta_j \gamma_j = \langle X - \mu, \psi\rangle_{L^2(\mathcal{T})}$, $\psi = \sum_{j=1}^n \theta_j \phi_j$, is univariate Gaussian. If X is mean-square continuous (equivalently, its covariance is continuous, the setting of Theorem 5.29), the Riemann sums $S_m = \sum_i (X(t_i) - \mu(t_i))\psi(t_i)\,\Delta t_i$ converge to S in $L^2(\Omega)$; each is a finite linear combination of the $X(t_i)$, hence Gaussian by Definition 5.10, with characteristic function $\mathbb{E}[e^{\mathrm{i}rS_m}] = \exp(\mathrm{i}r\,\mathbb{E}[S_m] - \frac{1}{2}r^2\mathrm{Var}(S_m))$, and letting $m \to \infty$ (the means and variances converge) gives $\mathbb{E}[e^{\mathrm{i}rS}] = \exp(\mathrm{i}r\,\mathbb{E}[S] - \frac{1}{2}r^2\mathrm{Var}(S))$, so S is Gaussian. For a merely square-integrable covariance, a jointly measurable process satisfying Definition 5.10 induces a Gaussian measure on its path space, of which each $\gamma_j = \langle X - \mu, \phi_j\rangle$ is a measurable linear functional, hence Gaussian; that is, X is an H-valued Gaussian in the sense of Definition 4.38 (see, e.g., Bogachev, *Gaussian Measures*, 1998, Proposition 2.3.9), which returns to the first case.

p202, p203 In Theorem 5.28 and its proof, three corrections are needed. (i) The sums should run over j with $\nu_j > 0$ only: for $\nu_j = 0$, $\xi_j = \gamma_j/\sqrt{\nu_j}$ is undefined, and $\gamma_j = 0$ a.s. since $\mathrm{Var}\,\gamma_j = \nu_j$. (ii) The theorem concludes $\xi_j \sim \mathrm{N}(0,1)$ iid, but the proof concludes only "pairwise independent" from Lemma 4.33. The proof of Lemma 4.33 factorises the joint density and so gives (mutual) independence, which is what "iid" requires; both should read "independent". (iii) The bound $\|X_J(t,\omega)\|_{L^2(\mathcal{T})} \le \|X(t,\omega)\|_{L^2(\mathcal{T})}$ from Lemma 1.41 holds for $\mu = 0$ but need not hold otherwise. Instead, $X - X_J = (I - P_J)(X - \mu)$ gives $\|X - X_J\|_{L^2(\mathcal{T})} \le \|X - \mu\|_{L^2(\mathcal{T})}$, and Exercise 4.13(b) applies.

p204 The equality in (5.39) follows from the uniform-in-time Mercer convergence in Theorem 5.29.

p207 In Definition 5.31, the identity should read

$$\|X(t+h) - X(t)\|_{L^2(\Omega)} = \mathbb{E}\big[|X(t+h) - X(t)|^2\big]^{1/2} \to 0 \quad \text{as } h \to 0,$$

with the exponent $1/2$, as in Definition 5.33.

p209 Proposition 5.37 claims $|C_X(s,t) - C_I(s,t)| \le K\Delta t^2$ in (5.47), but its proof gives only first order: (5.48) is $\|X(t) - I(t)\|_{L^2(\Omega)} \le K\Delta t$, and the Cauchy–Schwarz step then gives $|C_X(s,t) - C_I(s,t)| \le K'\Delta t$. "The estimate (5.47) follows from (5.48)" is therefore false. Under mean-square differentiability alone, the second-order rate is not established. It does hold under the hypothesis of Corollary 5.38, $C_X \in \mathrm{C}^2(\mathcal{T}\times\mathcal{T})$ with $\mathcal{T}$ closed and bounded. Since $I(s) = \sum_a w_a(s)X(t_a)$ with hat-function weights, $C_I(s,t) = \sum_{a,b} w_a(s)w_b(t)C_X(t_a,t_b)$ is the tensor-product piecewise-linear interpolant of C_X. Interpolating first in s and then in t (the weights are convex) gives, uniformly in s,t,

$$|C_X(s,t) - C_I(s,t)| \le \frac{\Delta t^2}{8}\Big(\sup|\partial_s^2 C_X| + \sup|\partial_t^2 C_X|\Big).$$

Corollary 5.38 (p210) and the application on p240, where the covariance $\tilde{c}_R$ is smooth, are correct with this version.

p210 In the proof of Proposition 5.37, the display following the appeal to Definition 5.33 should use the difference quotient over $[t_j, t_{j+1}]$, matching the identity established at the start of the proof:

$$\frac{X(t) - X(t_j)}{t - t_j} - \frac{X(t_{j+1}) - X(t_j)}{t_{j+1} - t_j} \to \frac{dX(t)}{dt} - \frac{dX(t)}{dt} = 0 \quad \text{in } L^2(\Omega).$$

p211 In Example 5.41, the sentence should read: hence if $n > 1/(2H)$, Theorem 5.40 applies with $p = 2n$ and $r = 2nH - 1 > 0$. As printed, with $p = \lceil 1/H\rceil$ and $r = pH - 1$, one has $r = 0$ whenever $1/H$ is an integer (in particular, $r = 0$ for $H = 1/2$, the case invoked in Corollary 5.42), whereas (5.50) requires $r > 0$. Taking $p = 2n$ also matches the even moments supplied by (5.51).

p211 In Example 5.41, the mixed derivative of the fBm covariance has the wrong sign: it should read $\partial^2 C/\partial s\,\partial t = +H(2H-1)|t-s|^{2H-2}$. The conclusion is unaffected: the derivative is unbounded at $s = t$ for $H < 1$, $H \ne \frac{1}{2}$; for $H = \frac{1}{2}$, $C = \min\{s,t\}$ is not differentiable on the diagonal.

p213 In the proof of Theorem 5.40, the second branch of the definition of Y should read $(1-t)X(0) + t\,X(1)$ and not $X(0) + t\,X(1)$.

p215 In Exercise 5.9, the process should read $B(t) = b\,t/T + W(t) - \frac{t}{T}W(T)$. As printed, the centred part of $B(t) = b\,t/T + (1 - t/T)W(t)$ has covariance $(1 - s/T)(1 - t/T)\min\{s,t\} = s(T - s)(T - t)/T^2$ for $s \le t$, and not the Brownian bridge covariance $\min\{s,t\} - st/T = s(T - t)/T$ of Lemma 5.22; the two agree only at $s = 0$.

p215 In Exercise 5.10, fBm $B^H(t)$ has long-range dependence for $H > 1/2$, not $H < 1/2$. With $\Delta(n) = B^H(n) - B^H(n-1)$,

$$\sum_{n=1}^{K} \mathbb{E}\big[\Delta(1)\Delta(n)\big] = \frac{1}{2} + \frac{1}{2}\big(K^{2H} - (K-1)^{2H}\big),$$

which converges to $1/2$ as $K \to \infty$ for $H < 1/2$ and diverges for $H > 1/2$.

Chapter 6. **p217** In the chapter introduction, the stationary process obtained as $t \to \infty$ from

$$\frac{du}{dt} = -\lambda u + \sqrt{2}\,\zeta(t)$$

has covariance $C(s,t) = \frac{1}{\lambda}\mathrm{e}^{-\lambda|s-t|}$. The stated covariance $C(s,t) = \mathrm{e}^{-|s-t|}$ holds only for $\lambda = 1$.

p218 In Definition 6.2, c is positive definite if the inequality (6.1) is strict for *distinct* $t_1, \ldots, t_n$ and $a_1, \ldots, a_n$ *not all zero*. As printed (strict for $a_1, \ldots, a_n \ne 0$, with repeated t_i allowed) no function is positive definite: $t_1 = t_2$, $a_1 = -a_2 = 1$ gives zero.

p221 In Example 6.8, the special case $q = 1/2$ of the Whittle–Matérn covariance should read $c_{1/2}(t) = \mathrm{e}^{-|t|}$, in agreement with (6.5), which defines c_q in terms of $|t|$.

p221 The identity (6.7) requires $\nu \ne 0$. For a stationary process of mean μ, the constant part contributes $\mu^2\big|\int_0^T \mathrm{e}^{-\mathrm{i}\nu t}\,dt\big|^2/(2\pi T)$, which is bounded by $2\mu^2/(\pi T\nu^2)$ and so vanishes as $T \to \infty$ whenever $\nu \ne 0$; at $\nu = 0$ it equals $\mu^2 T/2\pi$ and diverges. Estimates of the spectral density from data are correspondingly useless at $\nu = 0$ unless the mean is subtracted first.

p221 In Example 6.9, the approximation of the spectral density should read

$$f_T(\nu_k) \approx \frac{T}{2\pi}|U_k|^2, \qquad \text{for } k = -J/2 + 1, \ldots, J/2,$$

with the factor T and the modulus. The identity quoted immediately above should read $f_T(\nu_k) = (T/2\pi)|u_k|^2$, and U_k is the trapezium rule approximation to u_k with the same normalisation; Algorithm 6.1 (`spectral_density.m`) correctly computes $T|U_k|^2/2\pi$.

p222 In Example 6.10, taking the y_k real keeps only the cosine terms, and the process $X(t)$ has covariance

$$\mathrm{Cov}(X(s), X(t)) \approx \tfrac{1}{2}\big(c(s-t) + c(s+t)\big), \qquad s,t \in [0,T),$$

up to a reflection at T. This equals $c(t)$ when $s = 0$, which is the case shown in Figure 6.5, but it is close to $\frac{1}{2}c(s-t)$ away from the ends of the interval. A stationary process with covariance close to $c(s-t)$ on $[0,T]$ is obtained by adding to each cosine term an independent sine term $\sqrt{\eta_k}\,\xi_k' \sin(\nu_k t)/\sqrt{T/2}$, that is, by taking $a_k = b_k(\xi_k + \mathrm{i}\xi_k')$ with ξ_k' an independent copy of ξ_k.

p233 In Example 6.30, the factor 2 in front of the expectation should be deleted, so that the covariance reads

$$C_Z(s,t) = \mathbb{E}\big[Z(s)\overline{Z(t)}\big] = 2\int_{\mathbb{R}} \mathrm{e}^{\mathrm{i}(s-t)\nu}\frac{\ell}{\sqrt{4\pi}}\,\mathrm{e}^{-\nu^2\ell^2/4}\,d\nu = 2c(s-t),$$

as in Example 6.29 and as required by the definition of C_Z on p232. The final value $2c(s-t)$ is correct.

p239 In (6.34), the inner sum over k (the part evaluated by the FFT) is missing; it should read

$$\widetilde{Z}_R(t_n) = \sum_{m=0}^{M-1} \mathrm{e}^{\mathrm{i}(-R+m\Delta\nu)t_n} \sum_{k=0}^{N-1} \mathrm{e}^{2\pi\mathrm{i}kn/N}\sqrt{f(\nu_{kM+m})}\,\Delta W_{kM+m},$$

as computed by Algorithm 6.4.

p240 In the proof of Corollary 6.37, the display following the citation of Lemma 6.36 should read

$$|c(s-t) - \widetilde{c}_R(s-t)| \le K_1\Big(\frac{1}{M^2}\frac{1}{N^{2q}} + \frac{N}{M^4} + \frac{1}{N^{2q}}\Big), \qquad \forall\, s,t \in [0,T],$$

with c in place of c_Z: Lemma 6.36 bounds $|c(t) - \widetilde{c}_R(t)|$, and as printed the bound is false, since $|c_Z - \widetilde{c}_R| = |2c - \widetilde{c}_R| \to |c|$ as $\widetilde{c}_R \to c$. In the same proof, the linear interpolant $I(t)$ is that of $\Re\widetilde{Z}_R(t)$, which has covariance $\widetilde{c}_R(t)$ and not $\widetilde{c}_R(t)/2$, because $\widetilde{Z}_R(t)$ has covariance $2\widetilde{c}_R(t)$; and display (6.36) should read

$$\max_{j,k=0,\dots,P-1} \bigl|c(s_j - s_k) - C_I(s_j, s_k)\bigr| \le K_1 \frac{N}{M^4} + \frac{K_1 + K_2}{N^2},$$

consistent with the entries $c_Z(s_j - s_k)/2 = c(s_j - s_k)$ of C_{true}. The rate (6.35) is unaffected.

p241 At the start of §6.5, the vector of samples should read

$$\boldsymbol{X} = \bigl(X(t_0), X(t_1), \dots, X(t_{N-1})\bigr)^{\mathsf{T}}, \qquad t_n = n\Delta t.$$

p243 In Example 6.43, the first column of D is $[c_0, \dots, c_4]^{\mathsf{T}} = [1, 2, 3, 3, 2]^{\mathsf{T}}$, for which $c_4 = c_1$ and $c_3 = c_2$ (not $c_5 = c_1$ and $c_4 = c_2$).

p255 In Exercise 6.10, f should be non-negative, and only its values on $[0, R]$ enter: read $f\colon [0, R] \to \mathbb{R}^+$.

p256 In Exercise 6.18, the second identity should read

$$d_k = c_0 + c_P(-1)^{k-1} + 2\sum_{j=1}^{P-1} c_j \cos\Bigl(\frac{\pi j(k-1)}{P}\Bigr), \qquad k = 1, \dots, 2P;$$

that is, $c_{P/2}$ should be c_P and the upper limit $P/2 - 1$ should be $P - 1$. For a $2P \times 2P$ circulant, $\omega = \mathrm{e}^{\pi \mathrm{i}/P}$, so the self-paired term of the first sum is $j = P$, with $\omega^{P(k-1)} = (-1)^{k-1}$, and the conjugate pairs $\pm j$ run over $j = 1, \dots, P - 1$; note $\omega^{P/2} = \mathrm{i}$, so $c_{P/2}(-1)^{k-1}$ cannot arise. The stated range $k = 1, \dots, P$ gives only P of the $2P$ eigenvalues.

p256 Exercise 6.21(b) is ambiguous, and under its literal reading the answer is negative. Fixing $\Delta t = 1/(N-1)$ ties the sample points to $[0, 1]$, so increasing N refines the grid rather than lengthening the interval, and refining never helps: for $\ell = 1$, the minimal extension has $\rho(D_-) = 3.83, 7.70, 15.4, 30.9$ at $N = 100$, 200, 400, 800. Only $\ell = 0.1$, for which $[0, 1]$ already spans ten correlation lengths, gives a non-negative definite $\widetilde{C}$ at every N. The intended reading is to pad, that is, to extend the sampled interval at fixed spacing Δt; the minimal extension then becomes non-negative definite once the interval covers about five correlation lengths.

Chapter 7. **p259** In Example 7.7, the pdf of $u(\boldsymbol{x}) = W_1(x_1)W_2(x_2)$ should read

$$p(u) = \frac{1}{\pi\sqrt{x_1 x_2}}\, K_0\Bigl(\frac{|u|}{\sqrt{x_1 x_2}}\Bigr),$$

with $\sqrt{x_1 x_2}$ in place of $x_1 x_2$ in both positions. As printed, the density has variance $(x_1 x_2)^2$, whereas the example's own covariance computation gives $\mathbb{E}[u(\boldsymbol{x})^2] = x_1 x_2$.

p262 In Example 7.12, the spectral density of the separable exponential covariance should read

$$f(\boldsymbol{\nu}) = \prod_{i=1}^{d} \frac{\ell_i}{\pi\,(1 + \ell_i^2 \nu_i^2)},$$

as computed for each factor in Example 6.6, equation (6.4). At $\boldsymbol{\nu} = \boldsymbol{0}$ the printed formula gives $\prod_i 1/(\pi\ell_i)$, whereas $f(\boldsymbol{0}) = (2\pi)^{-d} \int_{\mathbb{R}^d} c(\boldsymbol{x})\, d\boldsymbol{x} = \prod_i \ell_i/\pi$; the two agree only when every $\ell_i = 1$.

p274 In Example 7.36, the definition of $\widetilde{C}_i$ should read $i = 0, 1$ and not $i = 1, 2$.

p278 In Example 7.40, the displayed matrix C_{red} is the one obtained for $a_{12} = -0.5$ in (7.32), not for the stated value: its $(1, 1)$ entry is $c(-1, -1) = \mathrm{e}^{-1} = 0.3679$ rather than $\mathrm{e}^{-3} = 0.0498$. For the stated a_{12}, the first and third columns of C_{red} are interchanged.

p279 The matrix V in the display at the top of the page has index typos. The first two entries of the second column should contain the indices $2n_1 + 1$ and $2n_1 + 2$ instead of n_2.

p285, p286, p289 In Example 7.45, the turning bands operator applied to the Bessel covariance should read

$$T_d\bigl(\Gamma(d/2)\, J_p(t)/(t/2)^p\bigr) = \cos(t), \qquad p = (d-2)/2,$$

so that $c_X(t) = \cos(t)$ and the factors $1/\sqrt{2}$ should be deleted from (7.34) and (7.35), giving $X(t) = \cos(t)\xi_1 + \sin(t)\xi_2$. By Definition 7.43 with $F^0(s) = \delta(s - 1)$, $T_d c^0(t) = \int_0^\infty \cos(st)\, dF^0(s) = \cos(t)$. With $c_X = \frac{1}{2}\cos$, Lemma 7.46 gives the turning bands field the covariance $\frac{1}{2\pi}\int_0^{2\pi} \frac{1}{2}\cos(r\cos\phi)\, d\phi = \frac{1}{2}J_0(r)$ when $d = 2$, half the Bessel covariance $c^0(r) = J_0(r)$ of Example 7.18, contradicting Proposition 7.47.

p286 In the proof of Proposition 7.47, the displayed Bessel identity should read

$$\frac{J_p(r)}{(r/2)^p} = \frac{2}{\pi^{1/2}\Gamma((d-1)/2)} \int_0^{\pi/2} \cos(r\cos\theta)\,\sin^{d-2}\theta\,d\theta, \qquad p = \frac{d-2}{2};$$

that is, a factor $r^{(d-2)/2}$ is missing from the printed coefficient $2^{2-d/2}/(\pi^{1/2}\Gamma((d-1)/2))$ (it is printed correctly in Appendix A.3).

p291 In the proof of Corollary 7.50, $|c^0 - \tilde{c}_M| = |c^0 - c_M| + |c_M - \tilde{c}_M|$ should read $|c^0 - \tilde{c}_M| \le |c^0 - c_M| + |c_M - \tilde{c}_M|$.

p305 In Lemma 7.67, the quantity $M_0(u)$ is infinite as stated and cannot be used for proving Theorem 7.68. We show here how to prove Theorem 7.68 using the Garsia–Rodemich–Rumsey (GRR) inequality. We state GRR on a bounded domain $D \subset \mathbb{R}^d$, rather than only on $D = (0,1)$, so that the argument covers Theorem 7.68 as stated. For an alternative approach to Theorem 7.68, see M. Talagrand, 'Upper and Lower Bounds for Stochastic Processes', Springer, 2021 (Section 2.5).

Let $\Psi(t) = \exp(t^2)$ and $p(t) = t^q$. (Note: GRR applies to a wider class of Ψ and p, which we do not describe here; see Garsia, Rodemich, and Rumsey (1971) for details. The multidimensional form stated here is due to Garsia, 'Continuity properties of Gaussian processes with multidimensional time parameter', Berkeley Symp. on Math. Statist. and Prob., 1972; for $d = 1$ and $D = (0,1)$ one may take $C_d = 8$ and $c_d = 4$.)

Lemma (Garsia–Rodemich–Rumsey)**.** *Let f be a continuous function on $\overline{D}$ and suppose that*

$$M_0(f) = \int_D \int_D \Psi\left(\frac{|f(\boldsymbol{x}) - f(\boldsymbol{y})|}{p(\|\boldsymbol{x}-\boldsymbol{y}\|_2)}\right) d\boldsymbol{x}\,d\boldsymbol{y} < \infty.$$

Then, for constants $C_d, c_d > 0$ depending only on d and D, we have, for all $\boldsymbol{x}, \boldsymbol{y} \in \overline{D}$,

$$|f(\boldsymbol{x}) - f(\boldsymbol{y})| \le C_d \int_0^{\|\boldsymbol{x}-\boldsymbol{y}\|_2} \Psi^{-1}\left(\frac{c_d M_0(f)}{r^{2d}}\right) dp(r).$$

The power r^{2d}, in place of the r^2 of the one-dimensional statement, arises because the estimate is localised at scale r to a pair of sets each of volume comparable with r^d; the factor C_d is the 8 of the classical one-dimensional inequality.

Lemma (Lemma 7.67 revised)**.** *Let $D \subset \mathbb{R}^d$ be as above, with $\operatorname{diam} D \le 1$. Fix $q > \epsilon > 0$. For a constant C_ϵ depending only on ϵ, q, d and D, we have, for any $u \in C(\overline{D})$,*

$$|u(\boldsymbol{x}) - u(\boldsymbol{y})| \le C_\epsilon \sqrt{\log(e + M_0(u))}\,\|\boldsymbol{x}-\boldsymbol{y}\|_2^{q-\epsilon}, \qquad \boldsymbol{x}, \boldsymbol{y} \in D, \tag{1}$$

where we assume

$$M_0(u) := \int_D \int_D \exp\left(\frac{|u(\boldsymbol{x}) - u(\boldsymbol{y})|^2}{\|\boldsymbol{x}-\boldsymbol{y}\|_2^{2q}}\right) d\boldsymbol{x}\,d\boldsymbol{y} < \infty. \tag{2}$$

Proof. With $\Psi(t) = \exp(t^2)$ and $p(t) = t^q$,

$$M_0(u) = \int_D \int_D \Psi\left(\frac{|u(\boldsymbol{x}) - u(\boldsymbol{y})|}{p(\|\boldsymbol{x}-\boldsymbol{y}\|_2)}\right) d\boldsymbol{x}\,d\boldsymbol{y},$$

which is finite by assumption, so GRR applies. Note that $\Psi(t) \ge 1$ for $t \ge 0$, that $\Psi^{-1}(t) = \sqrt{\log t}$ for $t \ge 1$, and that $p'(t) = q\,t^{q-1}$. Put $N := \max\{c_d M_0(u), e\}$, so that $\log N \ge 1$ and, for a constant $\kappa \ge 1$ depending only on c_d, $\log N \le \kappa \log(e + M_0(u))$. Since Ψ^{-1} is increasing and $r \le 1$,

$$\begin{aligned}
\sqrt{\log(c_d M_0(u)/r^{2d})} &\le \sqrt{\log N + 2d\,|\log r|} \\
&\le \sqrt{\log N} + \frac{d}{\sqrt{\log N}}|\log r| \\
&\le \sqrt{\log N} + \frac{d}{\epsilon\,e}\,r^{-\epsilon},
\end{aligned}$$

using $\log N \geq 1$, $\sqrt{a+b} \leq \sqrt{a} + \frac{b}{2\sqrt{a}}$ for $a, b > 0$, and $|\log r| \leq r^{-\epsilon}/\epsilon e$ for $r \in (0,1)$. Then, from GRR, for any $\boldsymbol{x}, \boldsymbol{y} \in D$,

$$\begin{aligned} |u(\boldsymbol{x}) - u(\boldsymbol{y})| &\leq C_d \int_0^{\|\boldsymbol{x}-\boldsymbol{y}\|_2} \sqrt{\log\left(\frac{c_d M_0(u)}{r^{2d}}\right)}\, q\, r^{q-1}\, dr \\ &\leq C_d \sqrt{\log N}\, \|\boldsymbol{x}-\boldsymbol{y}\|_2^q + \frac{C_d\, d}{\epsilon\, e}\, \frac{q}{q-\epsilon}\, \|\boldsymbol{x}-\boldsymbol{y}\|_2^{q-\epsilon}. \end{aligned}$$

As $\|\boldsymbol{x}-\boldsymbol{y}\|_2 \leq \operatorname{diam} D \leq 1$, the first term is bounded by $C_d\sqrt{\log N}\,\|\boldsymbol{x}-\boldsymbol{y}\|_2^{q-\epsilon}$, and $\sqrt{\log N} \leq \sqrt{\kappa}\sqrt{\log(e + M_0(u))}$, while $\sqrt{\log(e+M_0(u))} \geq 1$ absorbs the second term. This gives (1) for a suitable C_ϵ. □

This version of Lemma 7.67 can be used in the proof of Theorem 7.68.

Theorem (Theorem 7.68 revised)**.** *Let $D \subset \mathbb{R}^d$ be a bounded domain and $\{u(\boldsymbol{x}) : \boldsymbol{x} \in \overline{D}\}$ be a mean-zero Gaussian random field such that, for some $L, s > 0$,*

$$\mathbb{E}\big[|u(\boldsymbol{x}) - u(\boldsymbol{y})|^2\big] \leq L\|\boldsymbol{x}-\boldsymbol{y}\|_2^s, \qquad \forall \boldsymbol{x}, \boldsymbol{y} \in \overline{D}. \tag{3}$$

For any $p \geq 1$ and any $\epsilon \in (0, s)$, there exists a random variable K such that $e^K \in L^p(\Omega)$ and

$$|u(\boldsymbol{x}) - u(\boldsymbol{y})| \leq K(\omega)\|\boldsymbol{x}-\boldsymbol{y}\|_2^{(s-\epsilon)/2}, \qquad \forall \boldsymbol{x}, \boldsymbol{y} \in \overline{D}, \quad a.s. \tag{4}$$

Proof. We can find a truncated Karhunen–Loève expansion $u_J(\boldsymbol{x})$ such that $u_J(\cdot, \omega) \in C(\overline{D})$ for all $\omega \in \Omega$. Fix $p \geq 1$ and let

$$\Delta_J(\boldsymbol{x}, \boldsymbol{y}) := \frac{u_J(\boldsymbol{x}) - u_J(\boldsymbol{y})}{\sqrt{8\, L\|\boldsymbol{x}-\boldsymbol{y}\|_2^s}}, \qquad \boldsymbol{x}, \boldsymbol{y} \in \overline{D}.$$

The random variables in the Karhunen–Loève expansion are *iid* with mean zero. Hence $\mathbb{E}\big[\Delta_{J+1} \,\big|\, \Delta_J\big] = \Delta_J$. Further, the function $t \mapsto \exp(t^2)$ is convex and Jensen's inequality implies that $\mathbb{E}\Big[e^{\Delta_{J+1}^2} \,\Big|\, \Delta_J\Big] \geq e^{\Delta_J^2}$. That is, $e^{\Delta_J^2}$ is a submartingale and Doob's submartingale inequality gives

$$\mathbb{E}\left[\sup_{1 \leq j \leq J} e^{2\Delta_j(\boldsymbol{x},\boldsymbol{y})^2}\right] \leq 4\mathbb{E}\Big[e^{2\Delta_J(\boldsymbol{x},\boldsymbol{y})^2}\Big].$$

Note that $\mathbb{E}\big[\Delta_J(\boldsymbol{x},\boldsymbol{y})^2\big] \leq L\|\boldsymbol{x}-\boldsymbol{y}\|_2^s/8\, L\|\boldsymbol{x}-\boldsymbol{y}\|_2^s \leq 1/8$ for every $\boldsymbol{x}, \boldsymbol{y}$. Consequently, $\Delta_J \sim \mathrm{N}(0, \sigma^2)$ where the variance satisfies $\sigma^2 \leq 1/8$. As $1/2\sigma^2 - 2 \geq 1/4\sigma^2$,

$$\mathbb{E}\Big[e^{2\Delta_J(\boldsymbol{x},\boldsymbol{y})^2}\Big] = \frac{1}{\sqrt{2\pi\sigma^2}} \int_{\mathbb{R}} e^{2x^2} e^{-x^2/2\sigma^2}\, dx \leq \frac{1}{\sqrt{2\pi\sigma^2}} \int_{\mathbb{R}} e^{-x^2/4\sigma^2}\, dx = \sqrt{2}.$$

We conclude that

$$\mathbb{E}\left[\sup_{1 \leq j \leq J} e^{2\Delta_j(\boldsymbol{x},\boldsymbol{y})^2}\right] \leq 4\sqrt{2}, \qquad \forall \boldsymbol{x}, \boldsymbol{y} \in \overline{D}.$$

Let

$$M_1(u) := \int_D \int_D \sup_{J \in \mathbb{N}} e^{2\Delta_J(\boldsymbol{x},\boldsymbol{y})^2}\, d\boldsymbol{x}\, d\boldsymbol{y},$$

so that, by monotone convergence, $\mathbb{E}[M_1] \leq 4\sqrt{2}\ \mathrm{Leb}(D)^2 < \infty$ and $M_1 \in L^1(\Omega)$. Now

$$2\Delta_J(\boldsymbol{x},\boldsymbol{y})^2 = \frac{|u_J(\boldsymbol{x}) - u_J(\boldsymbol{y})|^2}{4L\,\|\boldsymbol{x}-\boldsymbol{y}\|_2^s} = \frac{|v_J(\boldsymbol{x}) - v_J(\boldsymbol{y})|^2}{\|\boldsymbol{x}-\boldsymbol{y}\|_2^{2q}}, \qquad v_J := \frac{u_J}{2\sqrt{L}}, \quad q := \frac{s}{2},$$

so that $M_0(v_J) \leq M_1(u)$ for every J. Applying Lemma 7.67 (revised) to v_J, with $\epsilon/2$ in place of ϵ, and multiplying through by $2\sqrt{L}$,

$$|u_J(\boldsymbol{x}, \omega) - u_J(\boldsymbol{y}, \omega)| \leq K(\omega)\|\boldsymbol{x}-\boldsymbol{y}\|_2^{(s-\epsilon)/2}, \qquad \forall \boldsymbol{x}, \boldsymbol{y} \in \overline{D},$$

for $K(\omega) = 2\sqrt{L}\,C_\epsilon \sqrt{\log(e + M_1(u))}$, which does not depend on J.

It remains to show $e^K \in L^p(\Omega)$. Put $a := 2\sqrt{L}\,C_\epsilon$. For $x \geq 0$ and $\delta > 0$ we have $\sqrt{x} \leq \delta x + 1/4\delta$, since $(\sqrt{\delta x} - 1/2\sqrt{\delta})^2 \geq 0$; taking $\delta = 1/pa$ and $x = \log(e + M_1(u))$ gives

$$p\,K = pa\sqrt{\log(e + M_1(u))} \leq \log(e + M_1(u)) + \frac{(pa)^2}{4}.$$

Hence $\mathbb{E}\left[e^{pK}\right] \leq e^{(pa)^2/4}(e + \mathbb{E}[M_1]) < \infty$, and $e^K \in L^p(\Omega)$.

As $u_J(\boldsymbol{x}) \to u(\boldsymbol{x})$ as $J \to \infty$ almost surely, this gives (4). □

p293 Theorem 7.52 needs the same corrections as Theorem 5.28 (see the entry for p202, p203): terms with $\nu_j = 0$, for which $\xi_j = \langle u - \mu, \phi_j\rangle/\sqrt{\nu_j}$ is undefined, are omitted from the sum (they vanish almost surely), and in the Gaussian case the ξ_j are independent, not only pairwise independent.

p297 In Theorem 7.57, V_d should be the volume of the unit ball in $\mathbb{R}^d$, namely $V_d = \pi^{d/2}/\Gamma(d/2 + 1)$, and not the volume of the unit sphere. The derivation of the theorem given in the Notes uses V_d in exactly this sense, via $\mathrm{Leb}(\{\boldsymbol{\nu} \in \mathbb{R}^d : \|\boldsymbol{\nu}\| < R\}) = V_d R^d$.

p299 In Theorem 7.61, the covariance should be cited as (7.52) and not (7.48): (7.48) is the one-dimensional covariance $\mathrm{e}^{-|x-y|/\ell}$ on $D = [-a, a]$, whereas the theorem concerns the separable exponential covariance with correlation lengths ℓ_1, ℓ_2 on $D = [-a_1, a_1] \times [-a_2, a_2]$.

p299 Theorem 7.61 should also be stated for $J \geq 2$. As printed it asserts $E_1 \leq 0$, since $\log^2 J/J$ vanishes at $J = 1$, whereas $E_1 > 0$ for a non-degenerate field. The proof requires $J \geq \mathrm{e}^2$ in any case, that being the range in which Exercise 7.17 is valid.

p300 Theorem 4.63 does not apply to (7.55), as C_J has rank at most $J < N$ and is singular (see p201). Since $C - C_J = \sum_{j>J} \nu_j \boldsymbol{\phi}_j \boldsymbol{\phi}_j^{\mathsf{T}}$ is non-negative definite, $\boldsymbol{u} = \boldsymbol{u}_J + \boldsymbol{Z}$ with $\boldsymbol{Z}$ independent of $\boldsymbol{u}_J$, and for ϕ with bounded second derivatives $|\mathbb{E}[\phi(\boldsymbol{u})] - \mathbb{E}[\phi(\boldsymbol{u}_J)]| \leq \frac{1}{2}\sup\|D^2\phi\|_2 \sum_{j>J}\nu_j\|\boldsymbol{\phi}_j\|_2^2$. Following (7.57), the closeness of averages for bounded continuous ϕ when C and $\widehat{C}_J$ are close holds because a Gaussian distribution depends continuously on its covariance matrix, not by Theorem 4.63.

p300 Following (7.57), the vector $\widehat{\boldsymbol{u}}_J$ has distribution $\mathrm{N}(\boldsymbol{\mu}, \widehat{C}_J)$ and not $\mathrm{N}(\boldsymbol{0}, \widehat{C}_J)$.

p303 In the display following (7.63), both occurrences of φ_j should read φ_k:

$$\sum_{k=1}^{P} a_k^j \int_D\int_D C(\boldsymbol{x}, \boldsymbol{y})\,\varphi_k(\boldsymbol{y})\varphi_i(\boldsymbol{x})\,d\boldsymbol{y}\,d\boldsymbol{x} = \hat{\nu}_j \sum_{k=1}^{P} a_k^j \int_D \varphi_k(\boldsymbol{x})\varphi_i(\boldsymbol{x})\,d\boldsymbol{x}.$$

p307 In the proof of Corollary 7.70, the exponent $(q - \epsilon)/2$ should read $(s - \epsilon)/2$ at all three occurrences, matching (7.74). Thus K is chosen with $\mathrm{e}^K \in L^q(\Omega)$ for $q = 2\lambda p\,|R|^{(s-\epsilon)/2} \geq 1$, and

$$\mathrm{e}^{\lambda p(u(\boldsymbol{x})-u(\boldsymbol{0}))} \leq \mathrm{e}^{\lambda p K(\omega)\|\boldsymbol{x}\|^{(s-\epsilon)/2}} \leq \mathrm{e}^{\lambda p K(\omega)|R|^{(s-\epsilon)/2}} = \mathrm{e}^{qK(\omega)/2}.$$

As written, this needs $\lambda > 0$ and $\boldsymbol{0} \in \overline{D}$, whereas the corollary is stated for all $\lambda \in \mathbb{R}$. Replace $\boldsymbol{0}$ by a fixed $\boldsymbol{x}_0 \in \overline{D}$ with $D \subset B(\boldsymbol{x}_0, R)$, λ by $|\lambda|$ in the exponents, and take $q = \max\{1, 2|\lambda|pR^{(s-\epsilon)/2}\}$; the final equality becomes $\leq$.

p310 In Exercise 7.1(a), the covariance should read $\mathbb{E}[W_N(\boldsymbol{x})W_N(\boldsymbol{y})] = \min\{x_1, y_1\}\min\{x_2, y_2\}$, and in Exercise 7.2(b) the distribution should read $\mathrm{N}(0,\, x_1x_2 + y_1y_2 - 2(x_1 \wedge y_1)(x_2 \wedge y_2))$. In both, the minimum is taken across the two points, as in the Brownian sheet covariance of Example 7.7.

p310 Exercise 7.2(c) is false: two-point differences $\beta(\boldsymbol{y}) - \beta(\boldsymbol{x})$ over opposite corners of disjoint rectangles need not be independent (for $[0,1]^2$ and $[2,3] \times [0,1]$, $\mathrm{Cov}(\beta(1,1), \beta(3,1) - \beta(2,0)) = 1$). It should concern the rectangular increments $\Delta_R\beta = \beta(b_1, b_2) - \beta(a_1, b_2) - \beta(b_1, a_2) + \beta(a_1, a_2)$, $R = [a_1, b_1] \times [a_2, b_2]$, which satisfy $\Delta_R\beta \sim \mathrm{N}(0, \mathrm{Leb}(R))$ and are independent for rectangles with disjoint interiors.

Chapter 8. **p316** In the caption of Figure 8.1, the parameter is $\sigma = 1/10$, not $\sigma = 1/2$: the horizontal lines are at $\pm 0.14 = \pm 2\sqrt{\sigma^2/(2\lambda)}$ for $\sigma = 1/10$, $\lambda = 1$. The histogram in (b) is of $P(t_n)$ for $t_n \in [500, 1000]$ (every time step, three paths), not $t_n \in [10, 1000]$.

p316 In Example 8.1, the limiting distribution $\mathrm{N}(0, k_{\mathrm{B}}T)$ of the momentum has pdf $p(P) = \mathrm{e}^{-\beta H(P)}/Z$ with $H(P) = P^2/2$. As a function of (Q, P), $\mathrm{e}^{-\beta P^2/2}$ cannot be normalised.

p316, p317 In the captions of Figures 8.1(a) and 8.2(b), the horizontal lines mark the 95% confidence interval and not the 90% interval. The interval $\big[-2\sqrt{\sigma^2/2\lambda},\ +2\sqrt{\sigma^2/2\lambda}\big]$ is ± 2 standard deviations of $\mathrm{N}(0, \sigma^2/2\lambda)$ and carries probability 0.954; the 90% interval is $\pm 1.645\sqrt{\sigma^2/2\lambda}$.

p318 In the caption of Figure 8.3, the volatility is $\sigma = 1$, not $\sigma = 5$.

p319 In §8.2, the σ-algebra should be the natural filtration $\mathcal{F}_s = \sigma(W(u) \colon u \le s)$ and not $\sigma(W(s))$. With $\mathcal{F}_s = \sigma(W(s))$, the identity $\mathbb{E}[W(t) \mid \mathcal{F}_s] = W(s \wedge t)$ fails for $t < s$: it gives $\mathbb{E}[W(t) \mid \sigma(W(s))] = (t/s)W(s)$, not $W(t)$ (the Brownian bridge).

p322 The identity (8.16) does not cover the case used in the proof of Lemma 8.9(iii), where $[r_k, r_{k+1}) \subset [s_j, s_{j+1})$. For $p \le s \le t$ and $p \le r$, it should read

$$\mathbb{E}\big[(W(r) - W(p))(W(t) - W(s)) \,\big|\, \mathcal{F}_s\big] = \max\{r \wedge t - s,\, 0\},$$

which includes both printed cases. The conclusion of the proof is unaffected.

p323 In Definition 8.12, the left-hand side of (8.18) should be squared,

$$\|I_n(t) - I_m(t)\|^2_{L^2(\Omega)} = \mathbb{E}\big[|I_n(t) - I_m(t)|^2\big] = \int_0^t \mathbb{E}\big[|X_n(s) - X_m(s)|^2\big]\, ds,$$

as in the Itô isometry (8.17).

p325 In the proof of Theorem 8.18, the random variable $\boldsymbol{X}$ should be $\mathcal{F}_0$-measurable, and hence independent of $\boldsymbol{W}(t)$, not "independent of $\mathcal{F}_0$". Independence of $\mathcal{F}_0$ does not give independence of $\boldsymbol{W}(t)$, and $\mathcal{J}(\boldsymbol{u})$ must be adapted.

p326 In (8.29), in the proof of Theorem 8.18, the constant should be L and not L^2: the linear growth bounds (8.24) of Assumption 8.17 carry the constant L. The same replacement is needed in the display from which (8.29) follows. The constant L^2 in the contraction estimate later in the proof is correct, as it comes from squaring the Lipschitz bound (8.25).

p328 In the proof of Lemma 8.19, the limit of $\mathbb{E}[B_N]$ should read

$$\to \frac{1}{2}\int_0^t \phi''(u(s))\, g(u(s))^2\, ds, \qquad \text{as } N \to \infty,$$

with g^2 in place of g, in agreement with $\mathbb{E}[C_j] = f^2\,\Delta t + g^2$ on the preceding line and with the g^2 in the statement (8.31).

p328 In Example 8.21, the parameters should be $\lambda > 0$ and $\mu, \sigma \in \mathbb{R}$. Mean reversion, the covariance and the limit $\mathrm{N}(\mu, \sigma^2/(2\lambda))$ all need $\lambda > 0$.

p332 In the expansion of $G(\boldsymbol{u}(r))$ in the derivation of the Milstein method, the remainder should be $\boldsymbol{R}_1(r; s, \boldsymbol{u}(s))$, not $\boldsymbol{R}_1(t, s; \boldsymbol{u}(s))$, as in the definition of $\boldsymbol{R}_M$ that follows.

p334 When the noise is diagonal, the reduction of Milstein's method to the componentwise form also needs g_{kk} to depend only on u_k. Otherwise the terms $\frac{\partial g_{kk}}{\partial u_i} g_{ii}$, $i \ne k$, remain, with products $\Delta W_{in}\Delta W_{kn}$ and Lévy areas $A_{ik,n}$. The commutativity condition should also carry the sum over ℓ: $\sum_\ell \frac{\partial g_{kj}}{\partial u_\ell} g_{\ell i} = \sum_\ell \frac{\partial g_{ki}}{\partial u_\ell} g_{\ell j}$.

p334 In §8.4, "A simple method for sampling the Lévy areas A_{ij} is considered in Exercise 5.3" should refer to Exercise 8.12, which identifies and approximates the Lévy area A_{12}. Exercise 5.3 concerns the Lévy–Ciesielski construction of Brownian motion.

p336 In Example 8.26, the exponent in the first line of the display for the second moment should be inside the expectation,

$$\mathbb{E}\big[u_n^2\big] = \mathbb{E}\Big[\Big(\prod_{j=0}^{n-1}\big(1 + r\,\Delta t + \sigma\,\Delta W_j\big)\Big)^2\Big] u_0^2 = \prod_{j=0}^{n-1} \mathbb{E}\big[(1 + r\,\Delta t + \sigma\,\Delta W_j)^2\big] u_0^2.$$

As printed, the left-hand side equals $\big(\mathbb{E}\big[\prod_j (1 + r\,\Delta t + \sigma\,\Delta W_j)\big]\big)^2 u_0^2 = (1 + r\,\Delta t)^{2n} u_0^2$, which is not $\mathbb{E}[u_n^2]$. The conclusion (8.52) is unaffected.

p339 In the proof of Lemma 8.30, the constant should be $2L$ and not $2L^2$, for the reason given for (8.29) on p326. Lemma 8.30 should also state the fourth-moment bound $\mathbb{E}\big[\|\boldsymbol{u}(t) - \boldsymbol{u}(s)\|_2^4\big] \le K|t - s|^2$, which the proof of Proposition 8.31(i) and (iii) needs for the second-order remainders: for example, $\|R_G\|_F \le \frac{1}{2} \sup \|D^2 G\|\, \|\boldsymbol{u}(r) - \boldsymbol{u}(s)\|_2^2$. It follows as for the second moment, with the Burkholder–Davis–Gundy inequality (p366) in place of the Itô isometry.

p340 In the proof of Proposition 8.31(ii), the integrand should read

$$\mathbb{E}\big[\|R_G(r;s,\boldsymbol{u}(s))\|_F^2\big] \quad \text{not} \quad \mathbb{E}\big[\|R_G(r;s,\boldsymbol{u}(r))\|_F^2\big].$$

The Itô isometry does not change the base point of the integrand.

p340 In the proof of Proposition 8.31(iii),

$$\mathbb{E}\big[\|DG(\boldsymbol{u}(s))\boldsymbol{R}_1(s,\boldsymbol{u}(s))\|_F^2\big] \quad \text{should read} \quad \mathbb{E}\big[\|DG(\boldsymbol{u}(t_j))\boldsymbol{R}_1(r;t_j,\boldsymbol{u}(t_j))\|_F^2\big],$$

matching the t_j-indexed sum and the R_G term alongside it. Likewise, $\boldsymbol{R}_1(s,\boldsymbol{u}(s))$ and $\boldsymbol{R}_1(t_j,\boldsymbol{u}(t_j))$ in the expansion of $\boldsymbol{R}_f$ that follows should read $\boldsymbol{R}_1(r;s,\boldsymbol{u}(s))$ and $\boldsymbol{R}_1(r;t_j,\boldsymbol{u}(t_j))$.

p340 In the proof of Proposition 8.31(iii), the sentence "using (8.43) gives for $r > s$." is incomplete. It should read "where the second equality uses (8.43) with $r > s$ to replace $\boldsymbol{u}(r) - \boldsymbol{u}(s)$."

p342 In the proof of Theorem 8.32, (8.61) should read $\boldsymbol{u}_{\Delta t}(t) - \boldsymbol{u}(t) = \int_0^t \boldsymbol{D}(s)\,ds + \int_0^t M(s)\,d\boldsymbol{W}(s) - \sum_{t_j<t} \boldsymbol{R}_M(t_{j+1}\wedge t;t_j,\boldsymbol{u}(t_j))$. The sum already contains the last partial interval, so the final term $\boldsymbol{R}_M(t;t_n,\boldsymbol{u}(t_n))$ should be deleted, and the sign of the sum is negative. The proof uses (8.61) only at grid times, where the deleted term vanishes, so the conclusion is unaffected.

p342 In the proof of Theorem 8.32, $\boldsymbol{e}$ is defined only at the grid times, by $\boldsymbol{e}(\hat{t}) = \boldsymbol{u}_{\Delta t}(\hat{t}) - \boldsymbol{u}(\hat{t})$, yet the bounds that follow are written $\mathbb{E}[\|\boldsymbol{e}(s)\|_2^2]$ for all s. Define instead $\boldsymbol{e}(t) \coloneqq \boldsymbol{u}_{\Delta t}(\hat{t}) - \boldsymbol{u}(\hat{t})$ for every $t \in [0,T]$. Then $\boldsymbol{e}$ is piecewise constant and $\mathcal{F}_{\hat{t}}$-measurable, with $\boldsymbol{e}(t_n) = \boldsymbol{u}_n - \boldsymbol{u}(t_n)$, and since $\boldsymbol{D}(s)$ and $M(s)$ depend on the approximation only through $\boldsymbol{u}_{\Delta t}(\hat{s}) - \boldsymbol{u}(\hat{s}) = \boldsymbol{e}(s)$, the displayed bounds hold as printed and Gronwall's inequality applies to $z(t) = \mathbb{E}[\|\boldsymbol{e}(t)\|_2^2]$.

p342 In the proof of Theorem 8.32, the bound $\mathbb{E}[\|M(s)\|_F^2] \le 2(L+L_2)\,\mathbb{E}[\|\boldsymbol{e}(s)\|_2^2]$ should read

$$\mathbb{E}\big[\|M(s)\|_F^2\big] \le 2\big(L^2 + L_2^2\, m\,(s-\hat{s})\big)\,\mathbb{E}\big[\|\boldsymbol{e}(s)\|_2^2\big] \le 2\big(L^2 + L_2^2\, m\,\Delta t\big)\,\mathbb{E}\big[\|\boldsymbol{e}(s)\|_2^2\big].$$

The Lipschitz conditions (8.25) and (8.59) are squared on taking second moments, and the second term carries the factor $\mathbb{E}\big[\|\int_{\hat{s}}^s d\boldsymbol{W}(r)\|_2^2\big] = (s-\hat{s})m$, which factorises out because the increment is independent of $\mathcal{F}_{\hat{s}}$ while $\boldsymbol{e}(s)$ is $\mathcal{F}_{\hat{s}}$-measurable. The existence of K_3, and hence the conclusion, is unaffected.

p345 In the caption of Figure 8.9, the coarse time step is $\Delta t = 0.02$ ($\kappa = 100$, as in Example 8.33), not $\Delta t = 0.04$.

p349 In Lemma 8.39, the bound should read $\mathbb{E}\big[\|D^n\boldsymbol{u}(t)(\boldsymbol{\xi}_1,\dots,\boldsymbol{\xi}_n)\|_2^2\big] \le K\|\boldsymbol{\xi}_1\|_2^2\cdots\|\boldsymbol{\xi}_n\|_2^2$, with squared norms on the right, as the proof (for $n=1$) gives and Corollary 8.41 uses.

p352 In the proof of Theorem 8.45, the coefficients of the frozen generator defined after (8.78) should be evaluated at the numerical solution $\boldsymbol{u}_{\Delta t}$ and not at $\boldsymbol{u}$:

$$\widehat{\mathcal{L}}(t)\Phi(t,\boldsymbol{x}) = \boldsymbol{f}(\boldsymbol{u}_{\Delta t}(\hat{t}))^{\mathsf{T}}\nabla\Phi(t,\boldsymbol{x}) + \frac{1}{2}\sum_{k=1}^{m} \boldsymbol{g}_k(\boldsymbol{u}_{\Delta t}(\hat{t}))^{\mathsf{T}}\nabla^2\Phi(t,\boldsymbol{x})\,\boldsymbol{g}_k(\boldsymbol{u}_{\Delta t}(\hat{t})),$$

and $\widehat{\mathcal{L}}^k(t)\Phi(t,\boldsymbol{x}) = \nabla\Phi(t,\boldsymbol{x})^{\mathsf{T}}\boldsymbol{g}_k(\boldsymbol{u}_{\Delta t}(\hat{t}))$. These are the coefficients of (8.77), to which the Itô formula is applied to give (8.78); compare Θ_1 in (8.81), which is evaluated at $\boldsymbol{u}_{\Delta t}(\hat{s})$.

p352 In the proof of Theorem 8.45, $e = \Phi(T,\boldsymbol{u}_0) - \Phi(0,\boldsymbol{u}_{\Delta t}(T))$ and the first line of (8.79) need an expectation on $\Phi(0,\boldsymbol{u}_{\Delta t}(T))$, and (8.79) should carry an overall minus sign,

$$e = -\mathbb{E}\Big[\int_0^T \frac{\partial}{\partial t}\Phi(T-s,\boldsymbol{u}_{\Delta t}(s)) + \widehat{\mathcal{L}}(s)\Phi(T-s,\boldsymbol{u}_{\Delta t}(s))\,ds\Big],$$

and consequently the display following (8.80) should read

$$e = \mathbb{E}\Big[\int_0^T \big(\mathcal{L}\Phi(T-s,\boldsymbol{u}_{\Delta t}(s)) - \widehat{\mathcal{L}}(s)\Phi(T-s,\boldsymbol{u}_{\Delta t}(s))\big)\,ds\Big].$$

For $d = m = 1$, $\boldsymbol{f} = \boldsymbol{0}$, $g(u) = u$ and $\phi(u) = u^2$, one has $\Phi(t,x) = x^2\mathrm{e}^t$ and $e = u_0^2\big(\mathrm{e}^T - (1+\Delta t)^N\big) > 0$, whereas the display as printed is negative. The theorem is unaffected, since (8.76) bounds $|e|$.

p353 In the proof of Theorem 8.45, $\widehat{\mathcal{L}}$ carries the factor $\frac{1}{2}$ but Θ_1, Θ_2 in (8.81)–(8.82) do not, so the decomposition of $\widehat{\mathcal{L}}\Phi - \mathcal{L}\Phi$ before them is out by a factor 2; define Θ_i with $\frac{1}{2}$. With the sign above, (8.84) then reads $e = -\int_0^T\int_{\hat{s}}^s \big(\mathbb{E}[Q_1] - \mathbb{E}[Q_2]\big)\,dr\,ds$.

p354 Below (8.88), "where $\boldsymbol{U}_\ell^{j,\mathrm{fine}}$ and $\boldsymbol{U}_\ell^{j,\mathrm{coarse}}$ are iid samples of $\boldsymbol{U}_\ell$" should read: for each j, $\boldsymbol{U}_\ell^{j,\mathrm{fine}}$ is a sample of $\boldsymbol{U}_\ell$ and $\boldsymbol{U}_{\ell-1}^{j,\mathrm{coarse}}$ is a sample of $\boldsymbol{U}_{\ell-1}$, the two being computed from a common Brownian path, and the pairs being independent over j. The coarse sample uses the time step $\Delta t_{\ell-1}$ and so is distributed as $\boldsymbol{U}_{\ell-1}$; and the pair is coupled, not independent, as the following sentence states. Were they iid samples of $\boldsymbol{U}_\ell$, then $\mathbb{E}[\mu_\ell] = 0$.

p361 In Theorem 8.51 (Wong–Zakai), the hypothesis $g(u) \geq g_0 > 0$ should be supplemented by an upper bound $g(u) \leq g_1$, and the two displays in the proof should read

$$|\phi(x) - \phi(y)| = \left| \int_x^y \frac{1}{g(s)}\, ds \right| \geq \frac{1}{g_1}|x - y|, \qquad |v(t) - v_J(t)| \leq g_1\, |\phi(v(t)) - \phi(v_J(t))|,$$

with g_1 in place of g_0. As printed, the hypothesis $g(u) \geq g_0$ gives $1/g \leq 1/g_0$ and hence the reverse inequality $|\phi(x) - \phi(y)| \leq |x - y|/g_0$; a lower bound on g alone does not make ϕ^{-1} Lipschitz.

p362 At the end of the proof of Theorem 8.51, the bound $\sup_t \left|\sum_{j>J} a_j \xi_j \sin((j + \frac{1}{2})\pi t)\right| \leq \left|\sum_{j>J} a_j \xi_j\right|$ is false (bounding $|\sin| \leq 1$ termwise gives $\sum_{j>J} a_j |\xi_j|$, which diverges). The uniform convergence of W_J to W almost surely is true, by the Itô–Nisio theorem (K. Itô and M. Nisio, Osaka J. Math. 5 (1968) 35–48). Alternatively, the proof applies unchanged to any piecewise continuously differentiable W_J with $W_J(0) = 0$ and $\sup_t |W(t) - W_J(t)| \to 0$ a.s.; the piecewise-linear interpolant of W at $t_j = j/J$ has this property because each path of W is uniformly continuous on $[0, 1]$.

p364 In the derivation of Heun's method, the Taylor remainder is $R = \frac{1}{2} g''(\zeta) g(v)^2 \delta^2$ for some ζ between v and $v + g(v)\delta$ (not $\frac{1}{2} g''(v) \xi^2 g(v)^2 \delta^2$), and with $\delta = \Delta W_n$ the remainder term in the following display is $-\frac{1}{2} R\, \Delta W_n$, not $+R$. Heun's method itself is unaffected.

p369 In Exercise 8.8(a), the limit should read

$$\mathbb{E}[u_n^2] \to \frac{\sigma^2}{2\lambda - \lambda^2\, \Delta t}, \qquad \text{as } n \to \infty,$$

with $2\lambda - \lambda^2\, \Delta t$ in the denominator. The Euler–Maruyama method (8.42) applied to (8.2) is $u_{n+1} = (1 - \lambda\, \Delta t) u_n + \sigma\, \Delta W_n$, so that $v_n = \mathbb{E}[u_n^2]$ obeys $v_{n+1} = (1 - \lambda\, \Delta t)^2 v_n + \sigma^2\, \Delta t$, with fixed point $\sigma^2\, \Delta t / \big(1 - (1 - \lambda\, \Delta t)^2\big)$. The limit exists for $\lambda\, \Delta t < 2$ and exceeds the exact value $\sigma^2/2\lambda$.

p369 In Exercise 8.10(a), $g(u_u)$ should read $g(u_n)$.

p369 In Exercise 8.10(b), the part is stated for $m = 1$, so there is a single Brownian motion and the increment driving component k should be ΔW_n, not $\Delta W_{k,n}$. The final term of the same display already carries ΔW_n^2.

p370 In Exercise 8.16(a), "$M = -1$" should read "$A = 1$". No M appears in the semilinear SODE (8.120), in the semi-implicit method (8.121) or in the exact solution (8.122); the linear coefficient is A, and M denotes the number of samples elsewhere in the chapter. The sign is fixed by (8.120), where the drift is written $-A\boldsymbol{u} + \boldsymbol{f}(\boldsymbol{u})$: taking $A = 1$ gives the decaying linear part that (8.122) describes, whereas $A = -1$ would give drift $+\boldsymbol{u} + \boldsymbol{f}(\boldsymbol{u})$.

p370 In (8.121), the trailing dt should be deleted. The step Δt already multiplies the drift, and the noise term carries the increment $\Delta \boldsymbol{W}_n$, so the equation reads

$$\boldsymbol{u}_{n+1} = \boldsymbol{u}_n + \Delta t\big(-A\boldsymbol{u}_{n+1} + \boldsymbol{f}(\boldsymbol{u}_n)\big) + G(\boldsymbol{u}_n)\, \Delta \boldsymbol{W}_n.$$

Chapter 9. **p372** In Example 9.1, "At each $x \in (0, 1)$, $a(x)$ is a uniform random variable with mean μ" should read "at each $x \in (0, 1)$, $a(x)$ has mean μ". In (9.4), $a(x)$ is a linear combination of P independent uniform random variables, which for $P \geq 2$ is not itself uniformly distributed: already for $P = 2$ the density of a sum of two independent, non-degenerate uniform random variables is trapezoidal and not constant.

p385 In (9.23), the bound should read

$$\mathbb{E}\big[E_{\mathrm{MC}}^2\big] \leq K^2 Q^{-1},$$

with K as defined in (9.24). The proof bounds $\mathbb{E}[E_{\mathrm{MC}}^2]$ by $Q^{-1} \|\tilde{u}_h\|^2_{L^2(\Omega, H_0^1(D))}$ and then applies Lemma 9.13 with $p = 2$, which gives $\|\tilde{u}_h\|_{L^2(\Omega, H_0^1(D))} \leq K$, so the square is required. Corollary 9.23 is correct as printed: its proof concludes $L = K^2$, which follows from $\mathbb{E}[E_{\mathrm{MC}}^2] \leq K^2 Q^{-1}$ but not from the printed KQ^{-1}.

p392 In the remark following Corollary 9.35, "If $d = 2$, we require $q > 4$" should read $q \geq 4$, consistent with the hypothesis $q \geq 2d$ of the corollary.

p393 In Example 9.36, the covariance of a is not close to c. It is $\frac{1}{2}\big(c_1(x-y) + c_1(x+y)\big)$, where $c_1(t) = \frac{1}{2} + \sum_{k\geq 1} \mathrm{e}^{-\pi k^2 \ell^2} \cos(k\pi t)$ is the covariance of the stationary process of Example 6.10 (see the entry for p222) with $T = 1$. For small ℓ, it is close to $\frac{1}{2}c(x-y)$ away from the ends of D. Likewise, in Example 9.37 the covariance is close to $\frac{1}{4}c^0$ away from the boundary of D. The eigenpairs, and the computations that use them, are unaffected.

p394 In the caption of Figure 9.9, the variance should read $\mathrm{Var}(a(\boldsymbol{x})) \approx 4.6708$ and not 8.0257. For the parameters stated in the caption ($\mu_k = 0$, $\sigma_k = 1$), the formula given in Example 9.39 yields $(\mathrm{e}^{\sigma_k^2} - 1)\,\mathrm{e}^{2\mu_k + \sigma_k^2} = \mathrm{e}^2 - \mathrm{e} \approx 4.6708$. The mean $\approx 1.7487 = 0.1 + \mathrm{e}^{1/2}$ in the same caption is correct.

p397 In the proof of Theorem 9.48, the three occurrences of u (twice in $|u(\cdot,\boldsymbol{y})|_{H^2(D)}$ and once in $|u|_{L^2_p(\Gamma,H^2(D))}$) should read $\tilde{u}$. Assumption 9.47 bounds $|\tilde{u}|_{L^2_p(\Gamma,H^2(D))}$, so the final step of the proof requires the tilde.

p397 Assumption 9.47 and Theorem 9.48 omit the boundary data. Theorem 9.48 is stated with $W^h \subset H^1_g(D)$, that is for non-homogeneous Dirichlet data, but Assumption 9.47 bounds $|\tilde{u}|_{L^2_p(\Gamma,H^2(D))}$ by a multiple of $\|\tilde{f}\|_{L^2_p(\Gamma,L^2(D))}$ alone, and so does (9.63). This is the point made for Assumption 2.64 in the p79 entry above: as printed, the assumption can be verified only for constant g. Both should carry a boundary term,

$$|\tilde{u}|_{L^2_p(\Gamma,H^2(D))} \leq K_2\Big(\|\tilde{f}\|_{L^2_p(\Gamma,L^2(D))} + \|g\|_{H^{3/2}(\partial D)}\Big),$$

and correspondingly, in (9.63),

$$|\tilde{u} - \tilde{u}_h|_W \leq K\sqrt{\tilde{a}_{\max}/\tilde{a}_{\min}}\; h\Big(\|\tilde{f}\|_{L^2_p(\Gamma,L^2(D))} + \|g\|_{H^{3/2}(\partial D)}\Big).$$

The intermediate step of the proof holds for any g, since $|\tilde{u}|_{H^2(D)}$ absorbs the boundary data; the term is lost only where Assumption 9.47 replaces $|\tilde{u}|_{H^2(D)}$ by $\|\tilde{f}\|$. For $g = 0$ the second term drops and the printed bounds are recovered.

p407 In Example 9.57, the MATLAB command should call `fem_blocks` and not `fem_components`. Algorithm 9.3 defines `[K_mats,KB_mats,f_vecs,wB]=fem_blocks(...)`, and the fifteen arguments supplied in the example match that signature.

p410 In Theorem 9.62, the conclusion that the eigenvalues of G_ℓ are the roots of $P_1, \ldots, P_{k+1}$ is correct, but the stated multiplicities are not: the count (9.106) is wrong except when $k + 1 - s \leq 2$, and in any case a multiplicity cannot be attached to the roots of each P_s separately, because different P_s share roots. For a symmetric weight, such as the Legendre or Hermite families, 0 is a root of every odd-degree P_s; already in Example 9.60 it is an eigenvalue of multiplicity 2. Nothing else in the chapter uses the multiplicities, and the two corollaries that follow need only the set of eigenvalues. For the multiplicities, see C. E. Powell and H. C. Elman, 'Block-diagonal preconditioning for spectral stochastic finite-element systems', IMA J. Numer. Anal. 29 (2009), 350–375. The proof also describes the permuted matrix as block tridiagonal; it is block diagonal.

p419 In (9.120), $\dfrac{dv^{k_i+1}}{dy^{k_i+1}}$ should read $\dfrac{d^{k_i+1}v}{dy_i^{k_i+1}}$.

p419 The order of differentiation is likewise misplaced in (9.121), in the statement of Lemma 9.75 and in (9.122), which should read $\dfrac{\partial^{k_i+1}v}{\partial y_i^{k_i+1}}$, $\dfrac{\partial^{k_i+1}\tilde{u}}{\partial y_i^{k_i+1}}$ and $\dfrac{\partial^{k_i+1}v(\boldsymbol{x},\cdot)}{\partial y_i^{k_i+1}}$ respectively. In (9.121) the subscript on y_i is also missing. The correct form appears in the final display of the proof of Lemma 9.75.

p423 In §9.6, "the Galerkin matrix $A \in \mathbb{R}^{(J+J_b)\times(J+J_b)}$" should read $A \in \mathbb{R}^{J\times J}$, in agreement with the Q linear systems "of dimension $J \times J$" in the same sentence. The matrices K_0 and K_ℓ occurring in $A = K_0 + \sum_{\ell=1}^{P} y_{r,\ell} K_\ell$ are indexed by $i, r = 1, \ldots, J$ in (9.93)–(9.94); the interior–boundary coupling is carried by the separate matrices $K_{\ell,\mathrm{B}} \in \mathbb{R}^{J_b\times J}$, which do not appear in this expression for A.

p424 In the Notes, "we introduced Assumption 9.47 as a generalisation of Assumption 2.13" should read "we introduced Assumptions 9.14 and 9.47 as generalisations of Assumption 2.64". Assumption 2.13 is the one-dimensional H^2 assumption; Assumption 2.64 is the two-dimensional one, to which the preceding remark on $\mathrm{C}^{0,1}$ coefficients refers. The next sentence should read "If Assumption 2.64 holds for almost every realisation, with a constant $K_2(\omega)$ that belongs to $L^4(\Omega)$, then Assumption 9.14 holds".

As printed, it names the wrong assumption (the $L^4(\Omega)$ condition is that of Assumption 9.14), and a pointwise H^2 condition with $|u|_{H^2(D)} \in L^4(\Omega)$ is not enough, since the assumptions bound $|u|_{H^2(D)}$ by a multiple of $\|f\|_{L^2(D)}$. In the same paragraph, "Theorem 9.17" should read "Lemma 9.17".

p425 In the Notes, the homogenised coefficient $a_* = 1/\mathbb{E}[a^{-1}]$ holds only for $d = 1$, and the homogenised PDE should read $-\nabla \cdot (a_* \nabla \bar{u}) = f$ with a_* a constant matrix. For $d \geq 2$ there is no closed formula: a_* is defined through the solution of a corrector problem on $\mathbb{R}^d$ and depends on the geometry of the realisations, not only on the distribution of a at a point. It lies between the harmonic mean $1/\mathbb{E}[a^{-1}]$ and the arithmetic mean $\mathbb{E}[a]$, and is a multiple of the identity when the law of a is isotropic.

p425 In the Notes, in the display defining the kernel $K(x, y)$ for the one-dimensional homogenisation corrector, the two cases should be interchanged:

$$K(x,y) \coloneqq \begin{cases} x(y-1/2), & 0 \leq x \leq y, \\ (x-1)(y-1/2), & y < x \leq 1, \end{cases}$$

equivalently $K(x,y) = (\frac{1}{2} - y)(\mathbf{1}_{y<x} - x)$. As printed, $K(0,y) = -(y - \frac{1}{2})$ and so $\int_0^1 K(0,y)^2\,dy = \frac{1}{12} \neq 0$, contradicting both $d(0) = 0$ and the variance stated immediately below. With the kernel above, $\int_0^1 K(x,y)^2\,dy = \frac{1}{12}\,x(1-x)(8x^2 - 8x + 3)$ exactly, reproducing the stated $\mathrm{Var}(d(x))$.

p426 In the Notes, "$[G_\ell]_{ij} = \langle \psi_\ell \psi_j, \psi_k \rangle_p$" should read $[G_\ell]_{jk} = \langle \psi_\ell \psi_j, \psi_k \rangle_p$.

p426 In the Notes, in the display following (9.130), u_{ik} should read u_{kj}. Setting $\boldsymbol{x}_* = \boldsymbol{x}_k$ and using $\phi_i(\boldsymbol{x}_k) = \delta_{ik}$ collapses the double sum in (9.130) to $\sum_{j=1}^{Q} u_{kj}\,\psi_j(\boldsymbol{y})$.

p428 In Exercise 9.6, $\|z\|_{L^2(\Omega, H_0^1(D))}$ should read $|z|_{L^2(\Omega, H^1(D))}$, since z need not vanish on ∂D, and "Corollary 9.17" should read "Lemma 9.17 and Corollary 9.18".

p429 Exercise 9.11 should also assume that Assumption 9.3 holds and that $f \in L^2(\Omega, \mathcal{F}, L^2(D))$. These give the existence and uniqueness of u and u^*, which the exercise takes for granted.

p430 In Exercise 9.20, "P_j, $i = 0, 1, \ldots$" should read "P_j, $j = 0, 1, \ldots$".

p430 In Exercise 9.23, "$\xi_k \sim \mathrm{U}(-\sqrt{3}, \sqrt{3})$ iid, $i = 1, \ldots, M$" should read "$\ldots$, $k = 1, \ldots, M$".

p430 In Exercise 9.24(a), $\|f(\cdot, \boldsymbol{y})\|_{L^2(D)}$ should read $\|\tilde{f}(\cdot, \boldsymbol{y})\|_{L^2(D)}$: the exercise is set for the approximate data $\tilde{a}$ and $\tilde{f}$, as in Theorem 9.76.

Chapter 10.

p432 In Example 10.1, "there are two spatially homogeneous solutions $u = \pm 1$ that are fixed points" omits $u = 0$. There are three spatially homogeneous fixed points: $u = \pm 1$, which are stable, and $u = 0$, which is unstable.

p435 In (10.9) and (10.10), $\frac{dY}{dx}$ should read $\frac{dZ}{dx}$, the derivative of the observation, and likewise in the sentence following (10.9), where it is $Z(x)$ that is only Hölder continuous. The SPDE samples the conditional distribution of the signal Y given the observations Z; the log-likelihood $\gamma^{-1} \int Y\,dZ - (2\gamma)^{-1} \int Y^2\,dx$ has gradient $\gamma^{-1}(dZ/dx - Y)$. As printed, the SPDE would need the signal it is meant to estimate. Figure 10.4 was computed with dZ/dx.

p435 The boundary condition at $x = 0$ should read $u(t, 0) = 0$, not $\frac{\partial u}{\partial x}(t, 0) = 0$, since the signal starts at $Y(0) = 0$.

p435 In the concrete example, $f(Y) = -Y$ and not $f(Y) = Y$. With $f(Y) = Y$ the signal grows exponentially, whereas Figure 10.4 shows the mean-reverting signal with $f(Y) = -Y$ on $[0, 50]$. (10.10) is unchanged, and the boundary condition at $x = b$ reads $\frac{\partial u}{\partial x}(t, b) = -u(t, b)$.

p438 In Example 10.8, the two eigenfunctions with the same eigenvalue should read $\sin(2\pi\ell x/a)$ and $\cos(2\pi\ell x/a)$, not $\sin(\ell x/a)$ and $\cos(\ell x/a)$: on $(0, a)$ the periodic eigenfunctions have period a, as the complex form $\mathrm{e}^{2\pi \mathrm{i} \ell x/a}$ in the same sentence shows.

p439 In Example 10.9, "$H_0^r(0, a)$ equals the fractional power space $\mathcal{D}(A^{r/2})$" holds for $1/2 < r < 3/2$, where the boundary values $u(0)$ and $u(a)$ are defined, but not in general; for instance, $\mathcal{D}(A) = H^2(0, a) \cap H_0^1(0, a) \neq H_0^2(0, a)$. The process of Examples 10.9 and 10.10 is $\mathcal{D}(A^{r/2})$-valued. The eigenfunctions should read

$$\chi_j(x) = \sqrt{2/a}\,\sin(j\pi x/a),$$

as in Example 10.10.

p440 In the caption of Algorithm 10.1, the value should read $\epsilon = 0.001$, to agree with `myeps=0.001` in the listing below it (and in `get_oned_bj.m`, Algorithm 10.3).

p441 In Example 10.11, "For $r = 1$ and $a = 1$" should read "For $r = 1$ and $a = 2\pi$": Figure 10.6 is drawn on $[0, 2\pi]$.

p441 In the caption of Algorithm 10.2, the first sentence should read "If `iFspace=0`, the output `dW` is a matrix of M columns with kth entry $W^{J-1}(t + \kappa\Delta t_{\mathrm{ref}}, x_k) - W^{J-1}(t, x_k)$ for $k = 1, \dots, J-1$." In the listing, `iFspace=1` returns the coefficients $b_j\xi_j$ and the `else` branch, `iFspace=0`, returns `dst1(X)`; Example 10.10 accordingly calls `get_onedD_dW(bj,kappa,0,1)` to generate the sample paths of Figure 10.5.

p441 In (10.18), both exponents should read $\mathrm{e}^{2\pi \mathrm{i}\, j(k-1)/J}$. With $x_k = a(k-1)/J$, the preceding line carries $\mathrm{e}^{2\pi \mathrm{i}\ell x_k/a} = \mathrm{e}^{2\pi \mathrm{i}\ell(k-1)/J}$; the printed $\mathrm{e}^{\mathrm{i}jk/J}$ drops the factor 2π and replaces $k-1$ by k. Only the corrected form is the inverse discrete Fourier transform named immediately below, as computed by `ifft` in `get_oned_dW.m`.

p442 In Example 10.12, $W^J(t, \boldsymbol{x})$ as displayed, with the complex eigenfunctions χ_{j_1,j_2} and real Brownian motions, is complex-valued. It should be the real-valued Q-Wiener process with these eigenvalues and eigenfunctions for $-J_i/2 < j_i \le J_i/2$, whose real eigenfunctions are the real and imaginary parts of the χ_{j_1,j_2}; this is what the construction that follows samples. In the same example, the eigenvalues of $-\Delta$ are $\lambda_{j_1,j_2} = 4\pi^2(j_1^2/a_1^2 + j_2^2/a_2^2)$, as Algorithm 10.5 uses; the printed $j_1^2 + j_2^2$ agrees with it only for $a_1 = a_2 = 2\pi$.

p444 In Definition 10.14, the cylindrical Wiener process is not U-valued: the series (10.20) does not converge in U, as the preceding paragraph says. It takes values in any Hilbert space $U_1 \supset U$ for which the inclusion is Hilbert–Schmidt.

p445 Before Theorem 10.16, "the L_0^2-valued process $B(t)$ is predictable (see Definition 8.6)" should refer to Definition 8.10: Definition 8.6 defines adapted processes, Definition 8.10 predictable ones.

p450 In the proof of Lemma 10.24, Mercer's theorem gives $\sum_j q_j\chi_j(\boldsymbol{x})^2 = q(\boldsymbol{x}, \boldsymbol{x})$ uniformly in $\boldsymbol{x}$, not $\sum_j q_j\|\chi_j\|_\infty^2 < \infty$. The former is what the proof needs:

$$\|G(u)\|_{L_0^2}^2 = \int_D g(u(\boldsymbol{x}))^2 q(\boldsymbol{x}, \boldsymbol{x})\, d\boldsymbol{x} \le \sup_{\boldsymbol{x}} q(\boldsymbol{x}, \boldsymbol{x})\, \|g(u)\|_{L^2(D)}^2.$$

p451 In the proof of Theorem 10.26, the contraction condition should read

$$T^2 + K^2 T^\zeta/\zeta < \frac{1}{2L^2},$$

and not $1/2L^2$. The preceding display gives $\|\mathcal{J}u_1 - \mathcal{J}u_2\|_{\mathcal{H}_{2,T}}^2 \le 2\left(T^2 + K^2T^\zeta/\zeta\right)L^2\|u_1 - u_2\|_{\mathcal{H}_{2,T}}^2$, so a contraction requires $2\left(T^2 + K^2T^\zeta/\zeta\right)L^2 < 1$.

p452 In Lemma 10.27, $\epsilon \in (0, \zeta)$ should read $\epsilon \in (0, \zeta/2)$; otherwise $\theta_1 = \min\{\zeta/2 - \epsilon, 1/2\}$ can be negative, and the bound on $A^{-\zeta/2+\epsilon}(I - \mathrm{e}^{-(t_2-t_1)A})$ used in the proof needs $\zeta/2 - \epsilon \ge 0$.

p453 In the proof of Lemma 10.27, the constant should read $C_{\mathrm{III}} = (C_1 + C_2)(1 + K_T)$, without the square. The bounds on III_1 and III_2 add to give $(C_1 + C_2)(t_2 - t_1)^{\theta_1}\left(1 + \sup_{0\le s\le T}\|u(s)\|_{L^2(\Omega,H)}\right)$, and (10.36) gives $1 + \sup_{0\le s\le T}\|u(s)\|_{L^2(\Omega,H)} \le (1 + K_T)\left(1 + \|u_0\|_{L^2(\Omega,H)}\right)$; the display that follows uses C_{III} linearly.

p453 In the proof of Lemma 10.27, the first line of the estimate of III_2' should read $\int_{t_1}^{t_2}$ in place of $\int_0^t$, since $\mathrm{III}_2' = \int_{t_1}^{t_2}\left(\mathrm{e}^{-A(t_2-s)} - I\right)G(u(s))\, dW(s)$ and the Itô isometry integrates over $[t_1, t_2]$, as on the line that follows.

p454 In the proof of Theorem 10.28, the integral $\int_0^t \|A^{1/2}\mathrm{e}^{-(t-s)A}\|_{\mathcal{L}(H)}^2\, ds$ is not finite: when the eigenvalues of A are unbounded, $\|A^{1/2}\mathrm{e}^{-sA}\|_{\mathcal{L}(H)}^2$ behaves like $(2\mathrm{e}s)^{-1}$ as $s \to 0$. The theorem holds: summing over the eigenfunctions χ_j of Q before integrating gives $\int_0^t \|A^{\zeta/2}\mathrm{e}^{-(t-s)A}\|_{L_0^2}^2\, ds \le \frac{1}{2}\|A^{(\zeta-1)/2}\|_{L_0^2}^2$. For the same reason, the first inequality of Exercise 10.8 should read $\int_0^T \|A^\beta \mathrm{e}^{-tA}v\|^2\, dt \le \frac{1}{2}\lambda_1^{2\beta-1}\|v\|^2$ for $v \in H$, with the operator norm removed.

p457 The covariance identity should read

$$\mathrm{Cov}\Big(W^J(t, x_i), W^J(t, x_k)\Big) = (t/h)\,\delta_{ik}, \qquad i, k = 1, \dots, J-1,$$

which is the index set used for $\boldsymbol{W}^J(t)$ immediately below. At $i = k = J$ the stated identity fails: $x_J = a$, so $W^J(t, x_J) = 0$ and the covariance is 0, not t/h.

p458 In Example 10.31, the three values quoted as the errors, 1.562×10^{-3}, 3.906×10^{-4} and 9.766×10^{-5} for $N = 64$, 256, 1024, are the time steps $\Delta t = T/N = 0.1/N$; the errors $|v(i) - 0.3489|$ from the computation displayed above should be reported in their place. As printed the values fall by exactly a factor 4 as N quadruples, so they fit Δt^1 and are inconsistent with both the quoted $0.96\,\Delta t^{0.65}$ and the order-$\Delta t^{1/2}$ rate stated next.

p461 In Theorem 10.34, "defined by (10.47)" should read "defined by (10.48)": the fully discrete approximation is (10.48), as the proof states.

p462 In the proof of Theorem 10.34, the factorisation in the estimate of III_1 should read

$$\left(\mathrm{e}^{-(t_n-s)A} - \mathrm{e}^{-(t_n-t_k)A}\right) A^{(1-\zeta)/2} = A^{1/2-\epsilon}\,\mathrm{e}^{-(t_n-s)A}\,A^{-\zeta/2+\epsilon}\left(I - \mathrm{e}^{-(s-t_k)A}\right),$$

and $\mathrm{e}^{-(t_n-t_k)A}$ should be replaced by $\mathrm{e}^{-(t_n-s)A}$ in the two integrals that follow. As printed, the factor $\mathrm{e}^{-(t_k-s)A}$ has $t_k - s < 0$ and is unbounded.

p465 In (10.54), the definition should read

$$\mathcal{G}(s;u) \coloneqq G(u) + G'(u)\,G(u)\int_{t_n}^{s} dW(r), \qquad t_n \le s < t_{n+1}.$$

The Milstein method (10.53) contributes $G(\tilde u_n)\Delta W_n + G'(\tilde u_n)G(\tilde u_n)I_{t_n,t_{n+1}}$, which the preceding display writes as $\int_{t_n}^{t_{n+1}}\left[G(\tilde u_n) + G'(\tilde u_n)G(\tilde u_n)\int_{t_n}^{s} dW(r)\right] dW(s)$; matching $\int_{t_n}^{t_{n+1}} \mathcal{G}(s;\tilde u_n)\,dW(s)$ in (10.48) forces the factor $G(u)$, and the proof of Lemma 10.36 uses the corrected form.

p466 In the proof of Lemma 10.36, the norm should read

$$\|(G(u(s)) - \mathcal{G}(s,u(t_k)))\,\chi\|_{L^2(\Omega,H)} \le \|R\|_{L^2(\Omega,H)}\|\chi\|_\infty,$$

as in the display that follows: for $\chi \in C(\overline{D})$ the quantity $(G(u(s)) - \mathcal{G}(s,u(t_k)))\,\chi$ takes values in H, not in $\mathcal{L}(H)$.

The bound on $R^\dagger$ needs a fourth moment: as $\|(u(s)-u(t_k))^2\|_{L^2(D)} = \|u(s)-u(t_k)\|^2_{L^4(D)}$,

$$\|R^\dagger(t_k,s)\|_{L^2(\Omega,L^2(D))} \le \tfrac12\|g''\|_\infty\,\mathbb{E}\Big[\|u(s)-u(t_k)\|^4_{L^4(D)}\Big]^{1/2},$$

which is not controlled by $\|u(s)-u(t_k)\|^2_{L^2(\Omega,H)}$ and (10.38). The lemma requires

$$\mathbb{E}\Big[\|u(s)-u(t)\|^4_{L^4(D)}\Big]^{1/4} \le K_4|s-t|^{\theta_2/2}$$

for $0 \le t \le s \le T$ (not needed when g is affine). This holds, for example, for $D \subset \mathbb{R}$; see Kruse, *Strong and Weak Approximation of Semilinear Stochastic Evolution Equations*, 2014, Theorem 2.31.

Mercer's theorem does not give $\sum_j q_j\|\chi_j\|^2_\infty < \infty$. It gives $\sum_j q_j\chi_j(x)^2 = q(x,x)$ uniformly on $\overline{D}$, so that

$$\|G(u(s)) - \mathcal{G}(s;u(t_k))\|^2_{L^2(\Omega,L^2_0)} = \mathbb{E}\left[\int_D R(x)^2\,q(x,x)\,dx\right] \le \sup_{x\in\overline{D}} q(x,x)\,\|R\|^2_{L^2(\Omega,H)}.$$

p467 In Corollary 10.38, Assumption 10.23 for $G = \sigma I$ reads

$$\|A^{(\zeta-1)/2}Q^{1/2}\|_{\mathrm{HS}(U,H)} < \infty \quad \text{not} \quad \|A^{(1-\zeta)/2}Q^{1/2}\|_{\mathrm{HS}(U,H)} < \infty.$$

Also, Theorem 10.34 gives the rate Δt^{θ_1} with $\theta_1 = \min\{\zeta/2-\epsilon, 1/2\}$, so (10.58) should read

$$\max_{0\le t_n\le T}\|u(t_n) - u_{J,n}\|_{L^2(\Omega,H)} \le K\Delta t^{\theta_1}:$$

for $\zeta > 1$ the drift term limits the rate to $1/2$ in this argument.

p468 In Example 10.39, for $r = 1$ ($\zeta = 2$) Corollary 10.38 guarantees the rate $1/2$ only. The observed rate 1 reflects a finer treatment of the drift term that is available for additive noise and smooth f.

p469, p470 Figure 10.11 (Example 10.40) was not computed with the stated $J_1 = J_2 = 128$, nor from the stated initial data. It used

$$u_0 = \sin(8x_1/\pi)\cos(\pi^2 x_2/64) \quad \text{not} \quad u_0 = \sin(x_1)\cos(\pi x_2/8),$$

and the former is not periodic on $(0, 2\pi) \times (0, 16)$.

p470 For the stochastic integral, the approximation is $\mathrm{e}^{-(t_{n+1}-s)\lambda_j} \approx \mathrm{e}^{-\Delta t \lambda_j}$ (the value at $s = t_n$), not $\mathrm{e}^{-t_{n+1}\lambda_j}$.

p471 In (10.62), the first term on the right-hand side should read $\mathrm{e}^{-\Delta t \lambda_j}\,\widehat{u}_{j,n}$, with the indices in the order j, n as on the left-hand side.

p473 In the paragraph preceding Algorithm 10.12, the sentence "As $J = J_w$, $G_h(\boldsymbol{u}_{h,n})$ is a diagonal matrix with entries $g(u^k_{h,n})$" should read: multiplication by g is diagonal (pointwise), and $G_h(\boldsymbol{u}_{h,n})\Delta \boldsymbol{W}_n$ is the $L^2(0,a)$ projection onto V^h of that product, with jth entry $\langle g(u_{h,n})\Delta W_n, \phi_j\rangle_{L^2(0,a)}$. By the definition given below (10.67), the j,k entry of $G_h(\boldsymbol{u}_{h,n})$ is $\langle G(\boldsymbol{u}_{h,n})\chi_k, \phi_j\rangle_{L^2(0,a)}$, which is not diagonal; Algorithm 10.12 accordingly assembles this term with the load-vector routine `oned_linear_FEM_b.m`.

p475 In Example 10.43, the observed errors of order h^2 are better than Theorem 10.41 guarantees, not "as predicted in Theorem 10.41 with $\zeta = 2$": with $J_w = J$ the condition $J_w \geq c\, h^{-\zeta/r}$ holds for $r = 1$ only when $\zeta \leq 1$, and the theorem then gives order h.

p480 In Exercise 10.4, $\mathrm{HS}(Q_1^{1/2}U_1 H)$ should read $\mathrm{HS}(Q_1^{1/2}U_1, H)$ (missing comma).

p480 In Exercise 10.7, the left-hand side of the second inequality should read $\mathbb{E}\big[\|AB\|^2_{L_0^2}\big]$; the expectation is taken twice in $\mathbb{E}\big[\|AB\|^2_{L^2(\Omega,L_0^2)}\big]$.

p480 In Exercise 10.8, the hypothesis should read $\beta \leq 0$, which is the range in which the bound is used in the proof of Theorem 10.26.

p480 In Exercise 10.12, the boundary condition should read $u(x) = \Delta u(x) = 0$ for $x \in \partial D$ (the variable x is missing).

Appendix.

p483 In §A.2, the trapezium rule on a single interval should read

$$\int_a^b u(x)\,dx \approx \frac{b-a}{2}\big(u(a) + u(b)\big),$$

and not $\frac{1}{2(b-a)}\big(u(a) + u(b)\big)$. This is the integral of the linear interpolant $p(x) = \big(u(a)(b-x) + u(b)(x-a)\big)/(b-a)$ given in the preceding line.

p487 In the proof of Corollary A.12, take $M(t) = |\beta(t)|$, which is a non-negative submartingale since $|\beta(s)| = |\mathbb{E}[\beta(t)\,|\,\mathcal{F}_s]| \leq \mathbb{E}[|\beta(t)|\,|\,\mathcal{F}_s]$; Theorem A.11 then gives (A.12). With $M(t) = |\beta(t)|^2$, as printed, Theorem A.11 bounds the fourth moment $\mathbb{E}[\sup_{0\leq t\leq T}|\beta(t)|^4]$ instead. Accordingly, the statement should read "$|\beta(t)|$ is a submartingale".

p487 The discrete Gronwall inequality (Lemma A.14) is incorrect. It should either be

Lemma 1. *Consider $z_n \geq 0$ such that $z_n \leq a + bz_{n-1}$ for $n = 1, 2, \ldots$ and $a, b \geq 0$. If $b = 1$, then $z_n \leq z_0 + na$. If $b \neq 1$, then*

$$z_n \leq b^n z_0 + \frac{a}{1-b}(1 - b^n).$$

or

Lemma 2. *Consider $z_n \geq 0$ such that*

$$z_n \leq a + b\sum_{k=0}^{n-1} z_k, \quad \text{for } n = 0, 1, 2\ldots \tag{*}$$

and constants $a, b \geq 0$. Then, $z_n \leq a(1+b)^n \leq a\exp(bn)$.

The first lemma is (Stuart and Humphries, 1997, Theorem 1.1.12).

To prove the second lemma, notice it is true for $n = 0$. Assume it is true for $z_0, \ldots, z_{n-1}$. Then,

$$\begin{aligned} z_n &\leq a + b \sum_{k=0}^{n-1} z_k \quad \text{by (*)} \\ &\leq a + b \sum_{k=0}^{n-1} a(1+b)^k \quad \text{by induction assumption} \\ &= a + ab \frac{1-(1+b)^n}{1-(1+b)} \quad \text{by geometric summation formula} \\ &\leq a + ab \frac{1-(1+b)^n}{-b} = a + ab((1+b)^n - 1) = a(1+b)^n. \end{aligned}$$

Therefore, $z_n \leq a(1+b)^n$ for all $n = 0, 1, 2, \ldots$ by induction. Finally, $z_n \leq a \exp(bn)$ as $1 + x \leq \exp(x)$ for $x \geq 0$. The second lemma can be used in the proof of Theorems 3.55 and 10.34.

p487–488 In §A.5, the value of the Euler constant should read "Its value is approximately 0.5772" and not 0.5572, as $\gamma = 0.57721566\ldots$